

```
import nbformat

# Load your notebook
nb = nbformat.read("C:/Users/werne/OneDrive/Documents/2024/Coding/Codecademy/Portfolio_Project

# Add metadata
nb.metadata.title = "Life Expectancy and GDP Analysis"
nb.metadata.authors = ["Werner Vorster"]

# Save it
nbformat.write(nb, "life_expectancy_analysis.ipynb")
```

Life Expectancy and GDP Analysis

Analyzing Global Health and Economic Indicators

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14 April 2025

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-

2. Data Preperation and Cleaning

2.1 Data Import, Cleaning, and Merging

2.1.1 Data Import

Issues with the Initial Dataset:

The initial dataset, sourced from Kaggle (uploaded by user KumarRajarshi), was found to contain incorrect data. Upon reviewing the comments from other users and conducting independent verification against official sources such as the World Bank, significant inconsistencies were identified. The population figures did not align with World Bank records, and further investigation revealed that all variables contained erroneous or unreliable data. Due to incorrect uploads or misused sources, the dataset was deemed unsuitable for analysis.

Importing Correct Data from Official Sources:

To ensure data accuracy, the necessary datasets were imported from three authoritative sources:

- **World Health Organization (WHO)**
- **World Bank (WB)**
- **Our World in Data (OWID)**

2.1 WHO Datasets

The following CSV files were downloaded from the WHO database and stored under `Datasets/Raw_datasets/WHO` :

- **Adult Mortality Rate**
- **Child Deaths per 1000 Births**
- **Infant Mortality**
- **Under-5 Mortality**

These datasets were read using Pandas' `.read_csv()` function and stored as variables.

2.1.1 Adult Mortality Dataset Processing

- Selected relevant columns: `'Location'`, `'Period'`, `'Dim1'`, `'FactValueNumeric'`
- Renamed columns:
 - `'Location'` → `'country'`
 - `'Period'` → `'year'`
 - `'Dim1'` → `'sex_classification'`
 - `'FactValueNumeric'` → `'adult_mortality'`
- Standardized data types (`.astype('string')`)
- Standardized country names to match the project template using `.replace()` :

- Cote d'Ivoire → Côte d'Ivoire
- Netherlands (Kingdom of the) → Netherlands
- North Macedonia → The former Yugoslav Republic of Macedonia
- Türkiye → Turkey

- Filtered the dataset to include only:
 - Data from 2000 to 2015
 - Entries labeled as "Both sexes"
- Dropped the `sex_classification` column

2.1.2 Child Deaths, Infant Mortality, and Under-5 Mortality Processing

- Selected relevant columns: `'Location'`, `'Period'`, `'Dim1'`, `'Dim2'`, `'Value'`
- Renamed columns:
 - `'Location'` → `'country'`
 - `'Period'` → `'year'`
 - `'Dim1'` → `'age_classification'`
 - `'Dim2'` → `'cause'`
 - `'Value'` → `'deaths'`
- Filtered and pivoted the datasets to obtain structured tables for analysis.

2.2 World Bank Datasets

The following CSV files were downloaded from the World Bank database and stored under `Datasets/Raw_datasets/World_Bank` :

- **Population**
- **GDP per Capita**
- **GNI per Capita**
- **Health Expenditure Percentage**
- **Life Expectancy**
- **Education Expenditure**
- **Unemployment Percentage**
- **Alcohol Consumption**
- **Government Health Expenditure**

These datasets were read using Pandas' `.read_csv()` function and stored as variables:

- `world_population` (loaded with `on_bad_lines='skip'` for data integrity)
- `wb_gdp` (GDP per capita)
- `wb_gni` (GNI per capita)
- `wb_health_exp_perc` (Health expenditure % of GDP)
- `world_life_expectancy` (Life expectancy at birth)
- `world_education_expenditure` (Education expenditure)
- `wb_world_unemployment_total` (Unemployment %)
- `world_alcohol_consumption` (Alcohol consumption per capita)
- `world_total_expenditure_gov` (Government health expenditure)

2.3 Our World in Data (OWID) Datasets

The following CSV files were downloaded from OWID and stored under `Datasets/Raw_datasets/Our_World_in_Data` :

- **BMI (Male and Female)**

These datasets were read and stored as:

- `world_in_data_bmi` (Male BMI)
- `world_in_data_bmi_female` (Female BMI)

3. Common Data Transformations

To standardize the imported datasets, the following transformations were applied across all datasets:

- **Column Renaming:** Standardized column names to match project conventions.
 - **Year Filtering:** Restricted data to the 2000–2015 period.
 - **Country Name Standardization:** Aligned country names with the project template.
 - **Data Type Conversion:** Ensured categorical variables were stored as strings.
 - **Sorting & Indexing:** Sorted by country and year for consistency.
-

2.1.2 Updating Template

To ensure consistency and usability throughout the project, the dataset template was refined through column renaming, data type standardization, and the removal of incomplete country records. These modifications enhanced the dataset's structure, making it more accessible for analysis.

Renaming Columns for Readability:

The dataset's original column names contained inconsistent capitalization, spaces, and special characters, which could hinder efficient data manipulation. To improve clarity and maintain a uniform naming convention, all column names were transformed to lowercase with words connected by underscores.

Methodology:

- The `.columns` method was used to inspect the dataset's variable names.
- The `.rename(columns={})` method was applied to systematically rename variables.
- **Example transformation:**

```
life_expectancy.rename(columns={'Country': 'country', 'Year': 'year'},  
inplace=True)
```

- This change ensured that variable names were standardized for consistency in all future operations.

Standardizing Country Data Type

The country variable was initially stored as an object type with inconsistent formatting. To maintain uniformity across datasets, it was explicitly converted to a string format.

- Transformation applied:

```
life_expectancy['country'] = life_expectancy['country'].astype('string')
```

- This standardization facilitated smooth dataset merging and prevented potential mismatches during analysis.

Removing Incomplete Country Entries

Some countries in the dataset had records for only a limited number of years, leading to inconsistencies in time series analysis. These incomplete entries were identified and removed through the following process:

1. The `.value_counts()` method was used to determine the frequency of each country in the dataset.
2. The result was converted into a dictionary where country names served as keys and their respective entry counts as values.
3. A list comprehension was applied to extract only countries with more than one entry:

```
valid_countries = np.array([key for key, value in unique_countries.items() if value > 1])
```

4. The length of the resulting array was printed using the `len()` function to verify the number of valid countries.
 5. A lambda function was used to scan through the dataset, retaining only entries present in `valid_countries`, while the rest were replaced with NaN values.
 6. Finally, rows containing NaN values in the country column were dropped using the `.dropna(subset=['country'])` method to ensure that only countries with complete records remained.
-

2.1.3 Data Cleaning and Merging

This section outlines the data cleaning and merging processes for the three main data sources: World Bank, WHO, and Our World in Data. The objective was to ensure consistency across datasets, resolve naming discrepancies, and integrate the data into a unified structure.

1. World Bank Datasets

Selecting Relevant Data

- Data from the World Bank was filtered to include only years 2000 to 2015.
- A list of countries from the World Bank dataset (`country_list_wb`) and a corresponding list from the life expectancy template (`country_list_le`) were created to identify common and mismatched country names.

Country Name Alignment

- Countries with mismatched names were identified by comparing `country_list_le` to `country_list_wb`.
- The `extractOne` function from the `thefuzz` library was used to match similar country names, with a score cutoff of 80.
- Manual adjustments were made for unmatched countries (e.g., "Korea, Dem. People's Rep.").

- The corrected names were applied to a cleaned version of the World Bank dataset (world_pop_cleaned).

Cross-Checking Countries

- A cross-checking list identified countries present in the life expectancy template but missing in the World Bank dataset (countries_cross_check).
- Missing countries such as Dominican Republic, Kyrgyzstan, and Swaziland were dropped from the Template.

Creating a Unified Dataset

- A new template (life_exp_clean) was created and sorted by country and year.
- The function modified_wb_pop_by_year() was developed to extract yearly data for population, GDP, health expenditure, and other variables.
- The cleaned and formatted World Bank dataset (wb_pop_sorted) was used to update life_exp_clean.
- This process was repeated for population, GDP, GNI, health expenditure, government health expenditure, life expectancy, unemployment, education expenditure, and alcohol consumption.

2. WHO Datasets

This section covers how data from the World Health Organization (WHO) was cleaned and merged into the main dataset. The WHO data primarily contributed to mortality statistics, including adult mortality, infant mortality, and under-5 mortality. WHO country names matched those in `life_exp_clean`, requiring no renaming.

2.1 Adult Mortality

- Countries absent from WHO data were identified and stored in `countries_not_included`.
- The `adult_mortality` column was updated using a left join with WHO data.
- Any missing values in `adult_mortality` were handled using the `where()` function, ensuring consistency.

2.2 Infant Mortality

- The dataset included mortality data for different causes of infant deaths. To standardize this, a new column, `infant_deaths`, was created by summing all causes of infant mortality for each country-year combination.
- This was done using a lambda function to apply the sum across relevant numeric columns.
- Any country that was not present in `life_exp_clean` was removed from the WHO dataset before merging.
- A merge operation replaced old values with WHO data, ensuring alignment across datasets.

2.3 Under-5 Mortality

- The process for under-5 mortality was identical to the steps for infant mortality:
 - All causes of under-5 deaths were summed into a new column.
 - Countries not found in `life_exp_clean` were dropped.
 - A merge operation replaced old values with WHO data to ensure alignment. The final dataset contained cleaned and integrated under-5 mortality data.

Key Takeaways:

- WHO data provided critical mortality statistics, integrated into `life_exp_clean`.
- Country names were already aligned, simplifying merging.
- Infant and under-5 mortality data required aggregation before merging.
- A left join ensured that all countries in the main dataset were preserved.

3. Our World in Data (OWID) Datasets?

3.1 BMI (Male & Female)

- Country names in OWID were checked against life_exp_clean.
- Similar to the World Bank process, extractOne was used to match country names.
- Manual corrections included:
 - "Lao People's Democratic Republic" → "Laos"
 - "Republic of Korea" → "South Korea"
 - "The former Yugoslav Republic of Macedonia" → "North Macedonia"
 - "United Kingdom of Great Britain and Northern Ireland" → "United Kingdom"
- The bmi_male and bmi_female columns were updated using cleaned OWID data.
- The Code column was dropped, and the dataset was sorted by country and year.

Final Adjustments and Verifications

- Sorting and Indexing: All datasets were sorted by country and year.
 - Data Integrity Checks: The number of unique countries was verified to be 176.
 - Standardization Across Sources: All variables were aligned with life_exp_clean, ensuring a unified and cleaned dataset for further analysis.
-

2.2 Filtering and Feature Selection

This section describes the process of filtering and selecting relevant features from the cleaned dataset to create a structured and standardized dataset for descriptive, exploratory, and inferential data analysis. This dataset forms the foundation for subsequent visualizations and statistical modeling.

1. Retained Variables

To ensure the dataset remains focused on key indicators relevant to life expectancy and economic conditions, the following variables were retained:

- country – Name of the country (176 unique countries)
- year – Observation year (2000 to 2015)
- life_expectancy – Probability of life expectancy at birth
- adult_mortality – Adult deaths per 1,000 adult population
- infant_deaths – Infant deaths per 1,000 infant population
- under_five_deaths – Under-five deaths per 1,000 under-five population
- bmi_male – Average BMI of the entire male population
- bmi_female – Average BMI of the entire female population
- health_expenditure_percentage – Expenditure on health as a percentage of Gross Domestic Product (GDP) per capita

- total_expenditure – General government expenditure on health as a percentage of total government expenditure
- population – Total population of the country
- gdp – Gross Domestic Product (GDP) per capita
- gni – Gross National Income (GNI) per capita

By filtering out unnecessary variables, the dataset remains streamlined for efficient analysis.

2. Filtering the Columns

To extract only the relevant variables, the following filtering process was applied:

```
life_exp_clean_2 = life_exp_clean.copy()
```

```
life_exp_clean_2 = life_exp_clean_2[  
    ['country', 'year', 'life_expectancy',  
     'adult_mortality', 'infant_deaths', 'under_five_deaths', 'bmi_male', 'bmi_female',  
     'health_expenditure_percentage', 'total_expenditure', 'population', 'gdp', 'gni']]
```

```
life_exp_clean_2.rename(columns={'gdp': 'gdp_per_capita'}, inplace=True)
```

This ensures that the final dataset includes only the variables necessary for analysis while renaming GDP to gdp_per_capita for clarity.

3. Converting Variables to Correct Data Types

To maintain consistency and accuracy in numerical computations, data types were adjusted as follows:

```
life_exp_clean_2[['year', 'population']] = life_exp_clean_2[['year',  
'population']].astype('Int64')  
float_columns = life_exp_clean_2.select_dtypes(include='Float64').  
columns  
life_exp_clean_2[float_columns] = life_exp_clean_2[float_columns].  
round(2)  
life_exp_clean_2['country'] = life_exp_clean_2['country'].  
astype('string')  
life_exp_clean_2['year'] = pd.to_datetime(life_exp_clean_2['year'],  
format='%Y').dt.year
```

- The year and population columns were converted to integers (Int64) to reflect their discrete nature.
- All floating-point numerical values were rounded to two decimal places to enhance readability.
- The country column was converted to a string type for categorical operations.
- The year column was briefly converted to a datetime format and then back to integer format to enable potential future time-series analysis.

These transformations ensure that the dataset is structured optimally for analysis, minimizing inconsistencies and formatting issues.

Final Adjustments and Verification

- Sorting and Indexing: The dataset was sorted by country and year to maintain chronological consistency.
- Data Integrity Checks: The total number of unique countries was verified to remain at 176.
- Standardization: All columns align with the predefined schema, ensuring uniformity across the dataset.

This structured approach results in a cleaned and well-formatted dataset, ready for further analysis and visualization.

2.3 Enriching the Data

This section describes the processes used to enrich the dataset by adding regional and sub-regional classifications and assigning income classes based on World Bank classifications. These enhancements provide additional layers of analysis, enabling a more comprehensive understanding of how geographic and economic factors relate to life expectancy.

1. Adding Regional and Sub-Regional Classifications

To enhance the dataset with regional classifications, all 176 countries included in the analysis were manually categorized into their respective regions and sub-regions. This classification was necessary to

facilitate regional comparisons and better contextualize variations in life expectancy and other key indicators.

Manual Categorization Process

- Country Listings: A complete list of 176 countries was reviewed, ensuring coverage across all geographic regions.
- Dictionaries for Sorting: Manually created dictionaries (`countries_by_region`) were developed to assign each country to its appropriate sub-region.
- DataFrame Creation: Separate sub-regional DataFrames were generated to store country classifications.
- Concatenation into Regional DataFrames:
 - Sub-regional DataFrames were concatenated using the `pd.concat()` function.
 - The final regional DataFrames were created using: `pd.concat(sub-regions).reset_index(drop=True).sort_values(by=['country'], ascending=True)`
- Exporting Cleaned Datasets:
 - The newly categorized data was exported using `.to_csv()` to create structured CSV files.
 - These files were saved in the project directory: `Datasets/Cleaned_datasets/World_regions`.

Integration into the Main Dataset

Two new categorical variables, region and sub-region, were added to the `life_exp_clean_2` DataFrame using a multi-loop classification system:

- A dictionary (`sub_regions_dict`) was created to map sub-regions to their corresponding regions.
- An iterative loop assigned each country to its respective region and sub-region.
- The classification was then stored in lists and appended to the main dataset.

The following code snippet illustrates the classification process:

```
# Generate lists of sub-regions
```

```
sub_region_keys = list(countries_by_region.keys())
```

```
sub_regions_dict = {  
    'Asian region': sub_region_keys[0:5],  
    'European region': sub_region_keys[5:9],  
    'African region': sub_region_keys[9:14],  
    'American region': sub_region_keys[14:18],  
    'Oceania region': sub_region_keys[18:22]  
}
```

```
# Assign region and sub-region classifications
```

```
region_classification_list = []
```

```
sub_region_classification_list = []
```

```
for country in list(life_exp_clean_2['country']):  
    for sub_region in countries_by_region:  
        if country in countries_by_region[sub_region]:  
            for regions in sub_regions_dict:  
                if sub_region in sub_regions_dict[regions]:  
                    region_classification_list.append(regions)  
                    sub_region_classification_list.append(sub_region)
```

```
# Add region and sub-region classifications as string-type columns
```

```
life_exp_clean_2['region'] = region_classification_list.astype('string')
```

```
life_exp_clean_2['sub_region'] = sub_region_classification_list.astype('string')
```

The inclusion of regional and sub-regional classifications allows for structured comparative analysis across different geographic areas.

2. Assigning Income Classes Based on World Bank Classifications

To further enrich the dataset, each country was assigned an income class based on the World Bank's classification thresholds. This classification aids in examining how economic factors impact life expectancy and related health indicators.

Income Classification Thresholds

The following World Bank income thresholds (2015 values) were applied:

- Low Income: GNI per capita $\leq$ 1,045 USD
- Lower-Middle Income: GNI per capita $\leq$ 4,125 USD
- Upper-Middle Income: GNI per capita $\leq$ 12,735 USD
- High Income: GNI per capita $>$ 12,735 USD

Automated Income Classification Process

To categorize each country, the following approach was implemented:

1. The income class was assigned based on each country's Gross National Income (GNI) per capita in 2015.
2. The classification was determined using a lambda function within a `apply()` method.
3. Missing values were forward-filled (`ffill()`) to ensure consistency.

The following Python code was used:

```
life_exp_clean_2['income_class'] = life_exp_clean_2['gni'][life_exp_clean_2['year']
== 2015].apply(
    lambda x: 'Low income' if x <= 1045 else
              'Lower-middle income' if x <= 4125 else
              'Upper-middle income' if x <= 12735 else
              'High income'
)
```

Forward-fill missing values

```
life_exp_clean_2['income_class'] = life_exp_clean_2['income_class'].ffill()
```

This process ensures that all 176 countries are correctly classified according to regional, sub-regional, and income groupings, enriching the dataset for advanced analysis and visualization.

2.4 Handling Missing Data

2.4.1 Inspection

To ensure data integrity and reliability in the analysis, a thorough inspection of missing values was conducted. The dataset was examined to identify missing values across all variables, including added categorical attributes such as region, sub-region, and income class.

Identification of Missing Data

A dictionary was generated to display all variables along with the corresponding count of missing entries. Additionally, a Missing Value Percentage Table was created to visualize the proportion of missing data per variable, highlighting areas that required imputation, extrapolation, or removal.

A key step in this inspection was identifying specific countries with missing values and quantifying these gaps. This was achieved by generating a dictionary of missing values per country, which facilitated a structured approach to addressing data gaps systematically.

Patterns in Missing Data

To categorize missing data trends, three distinct conditions were identified:

1. Countries missing values only for earlier years – These cases suggest that data collection for these countries commenced after a certain period.
2. Countries missing values only for later years – These cases indicate an abrupt cessation of data recording, possibly due to political instability, economic downturns, or shifts in reporting standards.
3. Countries missing all values – These cases highlight a complete lack of recorded data for specific variables, necessitating alternative handling strategies such as exclusion or alternative estimations.

The missing data distribution was further analyzed in the context of three key variables requiring imputation:

- Health Expenditure Percentage and Total Expenditure (Government Health Expenditure): Missing values were recorded for the same set of countries and years, namely Afghanistan, Iraq, Montenegro, Timor-Leste, Zimbabwe, Libya, and the Syrian Arab Republic.
- Gross National Income (GNI per capita): Missing values were primarily observed in earlier years for Afghanistan, Djibouti, and Nigeria.

Challenges in Data Imputation

- The dataset contained only 16 yearly observations per country per variable, limiting the training capability of the imputer.
- Due to insufficient data points, multiple imputation could only estimate a single fitting value rather than a range of plausible values for each missing entry.
- Economic fluctuations further complicated imputation, as drastic changes in macroeconomic variables were difficult to account for with limited data.
- In practice, this meant that for multiple imputation techniques, a single fitting value was often used to impute all missing data for a variable, reducing the robustness of the imputation model.

Testing Multiple Imputation Approaches

Several imputation strategies were explored to determine the most effective method:

- Imputation using regional-level aggregate data
- Imputation using sub-regional aggregate data
- Imputation using all non-missing data

Correlation Analysis and Insights

A seaborn heatmap of correlation matrices was employed to investigate potential relationships between missing variables and other recorded variables. The key findings were as follows:

1. **Health Expenditure Percentage:** Displayed weak correlations with other variables, limiting the effectiveness of predictive imputation. The highest correlations observed were:

- Life Expectancy: 0.36
- Adult Mortality: 0.22
- Infant Deaths: 0.36
- Under-5 Deaths: 0.28
- Male BMI: 0.36
- Female BMI: 0.15
- GDP per Capita: 0.35
- GNI per Capita: 0.41

2. **Total Expenditure (Government Health Expenditure):** Exhibited stronger correlations, suggesting potential for imputation based on related variables. Notable correlations included:

- Life Expectancy: 0.55
- Adult Mortality: 0.40
- Infant Deaths: 0.58
- Under-5 Deaths: 0.46
- Male BMI: 0.47
- Female BMI: 0.28
- GDP per Capita: 0.41
- GNI per Capita: 0.41

Summary of Missing Data Handling Methods

Based on the above analysis, three primary techniques were employed to handle missing data:

1. Imputation: Applied to variables where sufficient correlation with other variables enabled reasonable estimates, such as GNI using GDP per capita trends.
2. Extrapolation: Used where missing data followed an identifiable temporal trend, allowing for reasonable projection of missing values.

3. Removal: Countries with missing values across all years for a variable were excluded from the dataset, as no meaningful estimation could be performed.

These methodologies were applied accordingly in the following sections to address missing values for Health Expenditure Percentage, Total Expenditure (Government Health Expenditure), and GNI.

2.4.2 Creating Multiple Imputation Class

Introduction

Missing data poses a significant challenge in any analysis, particularly when working with economic and health indicators where data collection may not be consistently available across all regions and years. To address the missing values in the `health_expenditure_percentage` and `total_expenditure` variables, a Multiple Imputation Class was developed. This class leverages an advanced imputation technique by identifying countries with similar economic and health characteristics to serve as reference data for missing entries.

Identifying Missing Data Patterns

A detailed analysis of missing data patterns revealed that for `health_expenditure_percentage` and `total_expenditure`, five countries—Afghanistan, Iraq, Montenegro, Timor-Leste, and Zimbabwe—had missing data primarily in the earlier years (2000 onward). Given that data collection improved in later years, it was inferred that the missing data was likely missing at random (MAR) rather than completely missing without a pattern. This assumption supported the approach of using similar countries with complete data for imputation.

Exploring Different Imputation Approaches

Initially, multiple strategies were tested to train the `IterativeImputer`:

1. Using regional data (aggregating data within broader regions).
2. Using sub-regional data (a more localized approach).
3. Using the entire dataset for training the imputer.

None of these methods produced reliable results due to the limited number of observations per country (only 16 yearly data points). This limitation meant that a single fitted value was often used across all missing entries, reducing the accuracy and robustness of the imputation.

Final Imputation Strategy

The optimal method involved finding countries with similar characteristics (excluding the missing variable) and then training the imputer on these selected countries. This approach allowed for a more contextually relevant imputation by ensuring that missing values were filled using countries with comparable economic and health profiles.

The imputation class, `CountryImputer`, follows these steps:

1. Identify similar countries based on shared economic and health characteristics.
2. Train an imputer using data from these selected countries, excluding the variable being imputed.
3. Apply imputation to fill missing values for the target country.

This methodology accounted for structural similarities between countries rather than relying on arbitrary regional groupings.

Class Implementation and Functionality

Step 1: Finding Similar Countries

The `find_similar_countries` method identifies relevant reference countries using a tolerance-based filtering approach:

- A set of core variables (life_expectancy, adult_mortality, infant_deaths, under_five_deaths, bmi_male, bmi_female, gdp_per_capita, and gni) are considered.
- The method calculates maximum and minimum values for each variable within the target country.
- A tolerance margin is applied to these values to create a range within which other countries should fall.
- Countries whose variable values fall within this tolerance range are selected as reference data.

This method ensures that only countries with comparable economic and health characteristics are used in the imputation process.

Step 2: Imputing Missing Data

The `impute_country` method applies Iterative Imputation to the selected reference data:

- The data is first normalized (e.g., GDP per capita and life expectancy are scaled for consistency).
- Certain variables (e.g., adult_mortality, infant_deaths) are inverted since lower values correlate with higher economic and health indicators.
- The `IterativeImputer` is trained using the selected reference countries, excluding the missing variable.
- The trained imputer is used to predict missing values for the target country.

Addressing Economic Fluctuations

One of the key challenges encountered was the economic fluctuations over time, which could not always be accounted for with the limited available data. Since imputation relies on patterns within existing data, unpredictable shifts in economic trends (such as political instability or sudden growth spurts) could reduce accuracy. Despite this limitation, the chosen method produced more reliable estimates than other tested approaches.

Example Usage of the Imputation Class

To demonstrate the application of the CountryImputer class, consider the following example where Afghanistan's health_expenditure_percentage is imputed:

```
finder = CountryImputer(life_exp_clean_2, 'Afghanistan',  
                        'health_expenditure_percentage')  
considered_countries = finder.find_similar_countries(10, ['bmi_male',  
                                                         'total_expenditure', 'gni'])  
hep_multiple_imp = finder.impute_country(considered_countries)
```

In this case:

- A 10% tolerance was applied to find similar countries.
- Certain variables (bmi_male, total_expenditure, gni) were excluded to focus on economic and mortality-related indicators.
- The resulting imputation was applied only to Afghanistan for the missing years.

Outputs and Verification

Throughout the imputation process, several diagnostic outputs were generated to ensure transparency:

1. Max and Min values per variable for the target country.
2. Max and Min values with applied tolerance to identify similar countries.
3. List of selected reference countries used for imputation.
4. Years with missing data to track imputed entries.
5. Final imputed dataset preview to validate the results.

Key notes

The development of this Multiple Imputation Class provides a structured and adaptable approach to handling missing data in economic and health datasets. By leveraging similarities between countries, this method ensures that missing values are imputed in a contextually relevant manner, improving the integrity and completeness of the dataset. While economic fluctuations remain a challenge, the approach provides a significantly better alternative to traditional imputation methods, ensuring that meaningful analyses can be conducted without the bias introduced by arbitrary data removal or simplistic imputation techniques.

2.4.3 Filling the missing values:

2.4.3.1 Health Expenditure Percentage

Identifying Missing Values:

During the inspection of missing values in the health_expenditure_percentage variable, the following countries were found to have incomplete data:

```
{'Afghanistan': 2, 'Iraq': 3, 'Libya': 4, 'Montenegro': 11, 'Somalia': 16, 'Syrian  
Arab Republic': 3, 'Timor-Leste': 3, 'Zimbabwe': 10}
```

To address these missing values, a multiple imputation strategy was adopted using a custom-built CountryImputer class, which applies IterativeImputer from Scikit-learn under the hood. This allowed for imputation based on similar countries and similar features, maintaining realistic values consistent with each country's economic and health profile.

Countries Imputed Using CountryImputer

1. Afghanistan

- **Missing Years:** 2000, 2001
- **Similar countries:** Guinea-Bissau, Liberia, Malawi, Rwanda
- **Excluded Features:** bmi_male, total_expenditure, gni
- **Imputed Values:** {2000: 10.11, 2001: 10.11}

2. Iraq

- **Missing Years:** 2000, 2001, 2002
- **Similar countries:** Azerbaijan, Egypt, Kazakhstan, Mongolia, Morocco, etc.
- **Excluded Features:** bmi_male, total_expenditure, gni
- **Imputed Values:** {2000: 3.16, 2001: 3.16, 2002: 3.16}

3. Montenegro

- **Missing Years:** 2000–2010 (11 years)
- **Similar countries:** Bosnia and Herzegovina, Croatia, Serbia, Slovakia, etc.
- **Excluded Features:** bmi_male, total_expenditure, gni
- **Imputed Values:** All missing years were imputed with 7.23

4. Timor-Leste

- **Missing Years:** 2000–2002
- **Similar countries:** Cambodia, India, Indonesia, Myanmar, etc.
- **Excluded Features:** bmi_male, total_expenditure, gni
- **Imputed Values:** {2000: 4.41, 2001: 4.41, 2002: 4.41}

5. Zimbabwe

- **Missing Years:** 2000–2009 (10 years)
- **Similar countries:** Malawi, Mozambique, Rwanda, Uganda, Zambia
- **Excluded Features:** bmi_male, total_expenditure, gni
- **Imputed Values:** All missing years were imputed with 5.87

Each imputation followed the same methodology:

1. Identify similar countries using selected features correlated with healthcare investment.
2. Apply iterative imputation constrained by max-min bounds and a tolerance range derived from existing values.
3. Evaluate the resulting values to ensure consistency with country-specific trends.

Imputation Using Extrapolation: Libya and Syrian Arab Republic

For countries where health expenditure percentage data was missing after a few years of available records, extrapolation techniques were used to fill in the gaps. Two different regression approaches were applied depending on the trend characteristics observed.

1. **Libya:** Polynomial Regression Imputation

- **Model Evaluation:**
 - **R² Score:** 0.0816
 - **Mean Squared Error (MSE):** 0.538
 - The low R² score and high MSE suggest that a linear model is a poor fit, indicating a non-linear trend.
- **Trend Observation:**
 - A clear upward trend in health expenditure percentage was observed for all countries in the Northern African sub-region.
 - Despite potential political reasons for missing data, a second-degree polynomial regression was used to extrapolate values.
- **Implementation:**

```
poly_model = make_pipeline(PolynomialFeatures(degree=2), LinearRegression())
poly_model.fit(recorded_years, recorded_hep_values)
poly_predicted_values = poly_model.predict(missing_years)
```
- **Imputed Values:**
{2012: 4.03, 2013: 4.49, 2014: 5.03, 2015: 5.64}
 - These values were used to fill in the missing years from 2012 onward.

2. Syrian Arab Republic: Linear Regression Imputation

- **Model Evaluation:**
 - R^2 Score: 0.847
 - Mean Squared Error (MSE): 0.087
 - These metrics indicate that linear regression is a strong fit for this country's data.
- **Trend Observation:**
 - A linear relationship was evident in the scatterplot of recorded data.
- **Implementation:**

```
model = LinearRegression()  
model.fit(recorded_years, recorded_hep_values)  
missing_hep_values_predict = model.predict(missing_years)
```
- **Imputed Values:**
`{2013: 2.64, 2014: 2.45, 2015: 2.27}`

The linear trend suggests that the missing values likely follow the recorded pattern, possibly withheld due to political unrest.

Dropping Somalia from the Dataset

During the data cleaning and imputation process, Somalia was removed from the dataset due to a complete absence of health expenditure percentage data across all years (2000–2015). Unlike other countries with partial missing data, Somalia lacked any available records, making it impossible to apply standard imputation techniques such as interpolation or extrapolation.

Further research into Somalia's economic and healthcare reporting context confirmed that this was not an oversight. Somalia stands out as an exception among African nations, notably not adhering to the Abuja Declaration adopted by the African Union, which recommends allocating at least 15% of national budgets to health. Instead, estimates suggest Somalia's health expenditure is consistently below 2%, with no reliable figures officially reported by the World Health Organization (WHO) during the study period.

Given the absence of comparable data and the unique political and economic situation of the country, retaining Somalia in the dataset would introduce substantial bias and uncertainty. Therefore, Somalia was justifiably excluded from the project dataset to maintain the overall data integrity and focus on countries with meaningful and verifiable health expenditure data.

2.4.3.2 Total Expenditure (Government Health Expenditure)

Identifying Missing Values

Upon inspection of the `total_expenditure` column—representing government health expenditure as a percentage of total government spending—a total of eight countries were found to contain missing values. The missing value counts per country were:

```
{ 'Afghanistan': 2, 'Iraq': 3, 'Libya': 4, 'Montenegro': 11, 'Somalia': 16, 'Syrian Arab Republic': 3, 'Timor-Leste': 3, 'Zimbabwe': 10 }
```

The missing values occurred across two distinct patterns: countries with missing data in the earlier years (e.g., 2000–2005), and those with data gaps in the later years (e.g., post-2011). First we'll focus on countries with missing values primarily in the earlier years: Afghanistan, Iraq, Libya, Montenegro, Timor-Leste, and Zimbabwe.

Countries Imputed Using CountryImputer

1. Afghanistan

- **Missing Years:** 2000, 2001
- **Excluded Features:** ['bmi_male', 'bmi_female', 'gni']
- **Similar Countries:** Guinea-Bissau, Liberia, Malawi, Rwanda
- **Imputed Values:** {2000: 2.01, 2001: 2.01}

2. Iraq

- **Missing Years:** 2000–2002
- **Excluded Features:** ['bmi_male', 'gdp_per_capita', 'gni']
- **Similar Countries:** Azerbaijan, Egypt, Morocco, Tajikistan
- **Imputed Values:** {2000: 2.82, 2001: 2.82, 2002: 2.82}

3. Montenegro

- **Missing Years:** 2000–2010 (11 years total)
- **Excluded Features:** ['bmi_male', 'gni']
- **Similar Countries:** Bosnia and Herzegovina, Bulgaria, Chile, Croatia, Cuba, Czechia, Estonia, Hungary, Lebanon, Maldives, Montenegro, Poland, Serbia, Slovakia, The former Yugoslav republic of Macedonia
- **Imputed Values:** 2000–2010: 10.81% (consistent across years)

4. Timor-Leste

- **Missing Years:** 2000–2002
- **Excluded Features:** ['bmi_male', 'gni']
- **Similar Countries:** Indonesia, Nepal, Yemen, among others
- **Imputed Values:** 2000–2002: 2.86%

5. Zimbabwe

- **Missing Years:** 2000–2009
- **Excluded Features:** ['bmi_male', 'gdp_per_capita', 'gni']
- **Similar Countries:** Malawi, Rwanda, Tanzania, Uganda
- **Imputed Values:** 2000–2009: 8.18%

Imputation Using Extrapolation: Libya and Syrian Arab Republic

In the dataset, both Libya and the Syrian Arab Republic have missing values for government health expenditure (total_expenditure) from 2012 onward and 2013 onward, respectively. Unlike countries where data is missing in earlier years due to late adoption of data collection, the absence of values in these later years suggests a different mechanism — Not Missing at Random (NMAR). It is reasonable to assume that political instability and civil conflict, which escalated during the missing periods, influenced the availability and publication of official health expenditure data.

Given this context, extrapolation was used to estimate the missing values using either linear or polynomial regression, depending on the observed data trend. Visual inspection through scatter plots was conducted to determine the appropriate method for each country.

1. **Libya:** Polynomial Regression Imputation

- **Model Evaluation:**
 - R^2 Score: 0.308
 - Mean Squared Error (MSE): 0.817
 - The low R^2 score indicates a weak linear relationship — the linear model explains only about 30.8% of the variance in the observed data. The relatively high MSE further supports that a linear model is not a good fit for Libya's trend in government health expenditure.
- **Trend Observation:**
 - The scatter plot of Libya's recorded total_expenditure data showed a non-linear pattern, inconsistent with a straight-line trend.
 - While Libya's values fluctuated, neighboring North African countries displayed a clearer upward trend. This regional context, along with Libya's earlier trajectory, supported the use of polynomial regression to better capture the curved progression.
- **Implementation:**

```
poly_model = make_pipeline(PolynomialFeatures(degree=2), LinearRegression())
poly_model.fit(recorded_years, recorded_hep_values)
poly_predict_known_values = poly_model.predict(recorded_years)
poly_predicted_values = poly_model.predict(missing_years)
rounded_poly_predicted_values = np.round(poly_predicted_values, 2)
```
- **Imputed Values:**

```
{2012: 5.07, 2013: 5.23, 2014: 5.44, 2015: 5.7}
```

2. Syrian Arab Republic: Linear Regression Imputation

- **Model Evaluation:**

- R^2 Score: 0.416
- Mean Squared Error (MSE): 0.733
- The R^2 score indicates a moderate fit — the linear model explains approximately 41.6% of the variance. The MSE is slightly lower than Libya's, suggesting a more stable and consistent trend. Overall, the linear model was reasonably well-suited for Syria's expenditure trend.

- **Trend Observation:**

- The scatter plot revealed a clear and consistent downward linear trend in total_expenditure before 2013.
- This supported the use of linear regression to estimate missing values from 2013 to 2015, in line with the observed trajectory.

- **Implementation:**

```
model = LinearRegression()  
model.fit(recorded_years, recorded_hep_values)  
missing_te_values_predict = model.predict(missing_years)
```

- **Imputed Values:**

```
{2013: 4.87, 2014: 4.68, 2015: 4.49}
```

The linear trend suggests that the missing values likely follow the recorded pattern, possibly withheld due to political unrest.

2.4.3.3 GNI per Capita

Overview

During the inspection of the gni (Gross National Income per capita) column, missing values were identified in three countries. The following dictionary summarizes the number of missing values per country:

```
{'Afghanistan': 2, 'Djibouti': 15, 'Nigeria': 8}
```

Given that GNI per capita closely mirrors the trends of GDP per capita, both in economic reporting and statistical progression, it was deemed reasonable to use GDP per capita as a predictor in imputing the missing GNI values. This assumption allowed the use of Iterative Imputation methods that model missing data based on multivariate relations, specifically between GDP per capita and GNI per capita.

Imputation Approach

For all three countries, the IterativeImputer from Scikit-learn was used. This approach models each feature with missing values as a function of other features and iteratively refines the imputed values.

Since the relationship between GDP per capita and GNI per capita tends to be linear and stable over time, it served as an appropriate foundation for imputation.

1. Afghanistan

- **Imputation Logic:** Afghanistan had sufficient non-missing data points to train its imputer using only its own records.
- **Implementation:**
 - **Selected columns:** ['gdp_per_capita', 'gni']
 - Fit and transformed values using IterativeImputer
 - Rounded results to one decimal place
 - Replaced missing gni values in Afghanistan's records using the imputed output
- **Imputed Values:**
{2000: 131.8, 2001: 169.4}
- **Justification:** Since Afghanistan's dataset had only two missing values and a good quantity of existing records, the internal trend between GDP and GNI was expected to yield accurate predictions without requiring external regional data.

2. Djibouti

- **Imputation Logic:** Djibouti had 15 missing GNI values, which made internal training less reliable. Instead, the imputer was trained using the complete GDP and GNI data of all countries within the East African sub-region, to which Djibouti belongs.
- **Implementation:**
 - Subset of training data: East African countries with complete gdp_per_capita and gni
 - Trained IterativeImputer on this regional data
 - Transformed missing values for Djibouti
 - Replaced missing gni values using the second column of the imputer output
- **Imputed Values:**
{2000: 706.24, 2001: 710.99, 2002: 712.25, 2003: 733.81, 2004: 774.96, 2005: 812.68, 2006: 865.83, 2007: 935.69, 2008: 1082.37, 2009: 1113.92, 2010: 1175.42, 2011: 1267.24, 2012: 1359.86, 2013: 2021.37, 2014: 2153.82}
- **Justification:** Regional socioeconomic factors are often shared within geographical groupings. Thus, imputing based on sub-regional data provided a stronger model than relying solely on Djibouti's limited historical data.

3. Nigeria

- **Imputation Logic:** Nigeria had 8 missing values. Similar to Djibouti, the imputer was trained using complete GDP and GNI values from the West African sub-region, which reflects Nigeria's regional economic context.
- **Implementation:**
 - Subset of training data: West African countries with no missing gdp_per_capita or gni
 - Trained and applied IterativeImputer to predict Nigeria's missing gni values
 - Replaced missing values in Nigeria's dataset accordingly
- **Imputed Values:**
{2000: 537.7, 2001: 536.2, 2002: 695.6, 2003: 744.9, 2004: 935.7, 2005: 1174.4, 2006: 1546.5, 2007: 1754.2}
- **Justification:** Although Nigeria has a large economy, regional context still improves imputation by capturing the economic behavior of peer nations with shared development patterns.

2.5 Data Type Adjustment

To optimize both memory efficiency and semantic clarity in the dataset, appropriate data type conversions were performed. Specifically, categorical variables were explicitly cast to the category data type in pandas. This step is essential for large datasets as it reduces memory usage and improves performance in operations such as grouping, sorting, and filtering. Additionally, assigning ordered categories can help preserve logical hierarchies in later visualizations and statistical modeling.

2.5.1 Income Class

The `income_class` column represents a natural ordinal scale ranging from low to high-income categories. This variable was explicitly converted into an ordered categorical type, following the World Bank classification:

```
['Low income', 'Lower-middle income', 'Upper-middle income', 'High income']
```

By setting the `ordered=True` parameter, the column retains its inherent order, which is useful for ordered visualizations and analysis (e.g., boxplots segmented by income class).

2.5.2 Region

The `region` column, while not ordinal, represents discrete geopolitical zones (e.g., Asia, Africa, Europe). This was converted to an unordered categorical type, ensuring efficient grouping and summarization without implying any hierarchy.

2.5.3 Sub-region

Similar to `region`, the `sub_region` column was also cast to a categorical type. This preserves its use for stratified analysis while benefiting from optimized storage.

By converting these columns, the dataset is not only lighter in memory but is also more semantically aligned for downstream analysis and plotting tasks, especially in libraries such as `seaborn` and `matplotlib`.

2.5.4 Display dataset information

The cleaned Life Expectancy dataset contains 2,800 entries (rows) and 16 columns, all of which have no missing values, indicating that the dataset is fully preprocessed and analysis-ready.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2800 entries, 0 to 2799
Data columns (total 17 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Unnamed: 0                            2800 non-null   int64
1   country                               2800 non-null   object
2   year                                  2800 non-null   int64
3   life_expectancy                       2800 non-null   float64
4   adult_mortality                       2800 non-null   float64
5   infant_deaths                         2800 non-null   float64
6   under_five_deaths                    2800 non-null   float64
7   bmi_male                             2800 non-null   float64
8   bmi_female                           2800 non-null   float64
9   health_expenditure_percentage         2800 non-null   float64
10  total_expenditure                     2800 non-null   float64
11  population                            2800 non-null   int64
12  gdp_per_capita                        2800 non-null   float64
13  gni                                    2800 non-null   float64
14  region                                2800 non-null   object
15  sub_region                            2800 non-null   object
16  income_class                          2800 non-null   object
dtypes: float64(10), int64(3), object(4)
memory usage: 372.0+ KB
None
```

Data Types Overview

- Numeric Columns:
 - int32: 1 column (year)
 - int64: 1 column (population)
 - float64: 10 columns (life_expectancy, adult_mortality, infant_deaths, under_five_deaths, bmi_male, bmi_female, health_expenditure_percentage, total_expenditure, gdp_per_capita, gni)
- Categorical Columns:
 - category: 3 columns (region, sub_region, income_class)
- String Columns:
 - string: 1 column (country)

Memory Usage

The entire dataset uses approximately 285.6 KB of memory, which is notably efficient given the data volume. This optimization was aided by converting relevant columns (e.g., regions and income classes) to the category data type.

Column Completeness

All columns have 2,800 non-null entries, confirming successful handling of previously missing values through appropriate imputation and cleaning strategies.

2.6 Exporting Cleaned Dataset to CSV

With all cleaning, imputation, and formatting processes completed, the fully processed DataFrame (`life_exp_clean_2`) was saved to a CSV file for future access and reproducibility:

```
life_exp_clean_2.to_csv("../Datasets/Cleaned_datasets/Life_expectancy_cleaned_1.csv")
```

The file was exported to the `Cleaned_datasets` directory under the main `Datasets` folder. This clean, analysis-ready file serves as the finalized version of the dataset for use in visualization, modeling, and reporting tasks in subsequent phases of the project. Saving this cleaned version ensures that all data wrangling efforts are preserved and provides a clear separation between raw and prepared data sources.

3. Data Wrangling and Statistical Analysis

3.1 Descriptive Statistics (Variables)

Code Explanation: How Summary Statistics Were Generated

To analyze life expectancy trends over time, descriptive statistics were calculated for each year from 2000 to 2015. These statistics include the mean, maximum, minimum, median, standard deviation, and inter-quartile range (IQR). Below is a brief example of how each was computed:

- **Mean:** For each year, the average life expectancy was calculated using `np.mean()` after filtering the dataset by year.
`mean = np.mean(life_expectancy['life_expectancy'].where(life_expectancy.year == year))`
- **Maximum and Corresponding Country:** The highest life expectancy value per year was identified using `np.max()`, and the corresponding country was retrieved using boolean indexing.
`max = np.max(life_expectancy['life_expectancy'][life_expectancy.year == year])`
`max_country = life_expectancy['country'][(life_expectancy.year == year) & (life_expectancy['life_expectancy'] == max)].values[0]`
- **Minimum and Corresponding Country:** The lowest life expectancy per year was calculated with `np.min()`, and the corresponding country was identified similarly to the max.
`min = np.min(life_expectancy['life_expectancy'].where(life_expectancy.year == year))`
`min_country = life_expectancy['country'][(life_expectancy.year == year) & (life_expectancy['life_expectancy'] == min)].values[0]`
- **Median:** The median life expectancy was calculated using `np.median()` after removing missing values.
`median = np.median((life_expectancy['life_expectancy'].where(life_expectancy.year ==`

```
year)).dropna())
```

- **Standard Deviation:** The standard deviation, which quantifies variation or dispersion in life expectancy for each year, was calculated with `np.std()`.
`std = np.std((life_expectancy['life_expectancy'].where(life_expectancy.year == year)).dropna())`

- **Inter-quartile Range (IQR):** The IQR was computed as the difference between the 75th percentile (Q3) and 25th percentile (Q1), providing a robust measure of spread.

```
q3, q1 = np.percentile(life_expectancy['life_expectancy']
[life_expectancy['year'] == year], [75, 25])
iqr = q3 - q1
```

After computing each metric, the resulting dictionaries were converted into DataFrames and merged into a final summary dataset:

```
descriptive_stats_life_exp = le_all_means_df.merge(le_all_max_df, on='year')\
.merge(le_all_min_df, on='year')\
.merge(le_all_median_df, on='year')\
.merge(le_all_std_df, on='year')\
.merge(le_all_iqr_df, on='year')
```

3.1.1 Life Expectancy

Summary statistics were computed for each year between 2000 and 2015. See below tables containing descriptive statistics for all data and per region:

Global Data:

year	mean	max	max_countries	minimum	min_countries	median	std	25%	75%	iqr
2000	66.77	81.08	Japan	44.52	Malawi	69.75	9.76	60.6	73.96	13.37
2001	67.1	81.42	Japan	41.96	Zimbabwe	69.96	9.77	60.8	74.32	13.52
2002	67.38	81.69	Japan	44.56	Zimbabwe	70.44	9.64	61.25	74.56	13.31
2003	67.67	81.76	Japan	43.39	Zimbabwe	70.72	9.54	61.51	74.6	13.09
2004	68.03	82.03	Japan	43.71	Lesotho	70.86	9.45	61.7	75.19	13.49
2005	68.38	81.96	Japan	43.23	Lesotho	71.25	9.31	61.88	75.19	13.31
2006	68.76	82.32	Japan	42.91	Lesotho	71.2	9.15	62.15	75.6	13.44
2007	69.1	82.51	Japan	43.12	Lesotho	71.42	8.99	62.58	75.7	13.12
2008	69.43	82.59	Japan	43.57	Lesotho	71.66	8.87	63.16	76.06	12.9
2009	69.84	82.93	Japan	44.03	Lesotho	71.95	8.67	63.61	76.26	12.64
2010	70.14	82.84	Japan	45.6	Lesotho	72.37	8.67	64.17	76.46	12.3
2011	70.6	82.7	Switzerland	46.69	Lesotho	72.56	8.36	64.34	76.74	12.4
2012	70.88	83.1	Japan	47.84	Lesotho	72.61	8.18	64.68	76.99	12.31
2013	71.17	83.33	Japan	49.0	Lesotho	72.74	8.06	64.8	77.13	12.33
2014	71.45	83.59	Japan	50.03	Lesotho	72.97	8.0	65.02	77.44	12.42
2015	71.65	83.79	Japan	51.1	Lesotho	73.14	7.79	65.51	77.54	12.03

- **Consistent Increase Over Time:** The global mean life expectancy rose steadily from 66.77 years (2000) to 71.65 years (2015) — an increase of nearly 5 years over the 15-year period.
- **Shifting Distribution:** The median rose alongside the mean (from 69.75 to 73.14), indicating an upward shift across the population, not just at the extremes.
- **Narrowing Spread:** The standard deviation decreased from 9.76 in 2000 to 7.79 in 2015, suggesting convergence among countries and a reduction in disparity.
- **Extremes:**
 - Japan consistently recorded the highest life expectancy, peaking at 83.79 years in 2015.
 - Lesotho had the lowest values throughout much of the dataset, although its life expectancy also improved (from 43.71 in 2004 to 51.1 in 2015).

- **IQR Trends:** The interquartile range (IQR) slightly decreased from 13.37 to 12.03, indicating that even the middle 50% of countries became more similar in life expectancy.

These trends point toward global improvements in life expectancy and reduced inequality, though certain countries continue to lag significantly behind.

Africa

year	mean	max	max_countries	minimum	min_countries	median	std	25%	75%	iqr
2000	55.08	73.69	Tunisia	44.52	Malawi	52.7	8.06	49.36	58.43	9.07
2001	55.37	73.82	Tunisia	41.96	Zimbabwe	53.15	8.1	49.9	58.57	8.67
2002	55.75	74.06	Tunisia	44.56	Zimbabwe	53.82	7.85	50.6	58.96	8.36
2003	56.16	74.31	Tunisia	43.39	Zimbabwe	54.1	7.76	50.95	59.22	8.27
2004	56.72	74.49	Tunisia	43.71	Lesotho	55.26	7.7	51.57	59.89	8.33
2005	57.24	74.76	Tunisia	43.23	Lesotho	55.71	7.58	52.34	60.37	8.03
2006	57.86	74.92	Tunisia	42.91	Lesotho	56.14	7.43	53.46	61.06	7.6
2007	58.45	75.07	Tunisia	43.12	Lesotho	56.52	7.34	54.22	61.64	7.42
2008	59.0	75.18	Tunisia	43.57	Lesotho	57.36	7.15	54.81	61.91	7.1
2009	59.61	75.27	Tunisia	44.03	Lesotho	58.12	7.0	55.77	62.71	6.93
2010	60.21	75.42	Tunisia	45.6	Lesotho	59.16	6.75	56.44	62.98	6.55
2011	60.73	75.46	Tunisia	46.69	Lesotho	60.02	6.45	57.08	63.33	6.25
2012	61.36	75.55	Tunisia	47.84	Lesotho	60.36	6.39	57.65	63.74	6.08
2013	61.85	75.65	Tunisia	49.0	Lesotho	60.73	6.22	58.29	63.95	5.66
2014	62.27	75.65	Tunisia	50.03	Lesotho	61.27	6.13	58.85	64.27	5.42
2015	62.74	75.69	Tunisia	51.1	Lesotho	61.64	5.97	59.21	64.66	5.45

- **Consistent Increase Over Time:** The mean life expectancy in Africa rose from 55.08 years (2000) to 62.74 years (2015) — an improvement of 7.66 years over the 15-year period. This rate of increase is notably higher than the global average of around 5 years.
- **Top Performers:** Tunisia held the highest life expectancy across all years, peaking at 75.69 years in 2015.
- **Lowest Performers:** Initially, Malawi and Zimbabwe had the lowest values (as low as 41.96 in 2001). From 2004 onward, Lesotho became the country with the lowest life expectancy, though even Lesotho saw improvement from 43.71 (2004) to 51.10 (2015).
- **Median Growth:** The median life expectancy increased steadily from 52.70 (2000) to 61.64 (2015), showing improvement not just at the top, but across the continent.
- **Decreasing Spread and Inequality:** The standard deviation dropped from 8.06 (2000) to 5.97 (2015), and the IQR (Interquartile Range) declined from 9.07 to 5.45, indicating a reduction in inequality and a convergence in life expectancy across African countries.
- **Quartile Movement:**
 - 25th percentile rose from 49.36 to 59.21
 - 75th percentile rose from 58.43 to 64.66
 - This upward shift in both percentiles further confirms the continent-wide progress.

Contextual Notes:

- The improvements in the minimum values suggest significant progress in the most disadvantaged countries.
- The narrowing of standard deviation and IQR shows life expectancy gains are more equitably distributed in Africa than before, even if absolute values remain lower than global figures.

Asia

year	mean	max	max_countries	minimum	min_countries	median	std	25%	75%	iqr
2000	69.13	81.08	Japan	55.3	Afghanistan	70.36	5.96	64.89	73.47	8.58
2001	69.59	81.42	Japan	55.8	Afghanistan	70.79	5.87	65.16	73.81	8.65
2002	69.96	81.69	Japan	56.45	Afghanistan	71.02	5.76	65.39	73.87	8.48
2003	70.31	81.76	Japan	57.34	Afghanistan	71.37	5.68	65.64	74.14	8.5
2004	70.48	82.03	Japan	57.94	Afghanistan	71.37	5.64	65.75	74.33	8.58
2005	70.92	81.96	Japan	58.36	Afghanistan	71.84	5.58	65.86	74.73	8.87
2006	71.23	82.32	Japan	58.68	Afghanistan	71.99	5.53	66.36	74.94	8.58
2007	71.53	82.51	Japan	59.11	Afghanistan	72.33	5.51	66.71	75.19	8.48
2008	71.7	82.59	Japan	56.51	Myanmar	72.4	5.71	67.11	75.49	8.38
2009	72.16	82.93	Japan	60.36	Afghanistan	72.7	5.37	67.9	75.84	7.94
2010	72.55	82.84	Japan	60.85	Afghanistan	73.16	5.34	68.29	76.27	7.98
2011	72.83	82.59	Japan	61.42	Afghanistan	73.31	5.26	68.57	76.61	8.04
2012	72.99	83.1	Japan	61.92	Afghanistan	73.45	5.32	68.72	77.06	8.34
2013	73.22	83.33	Japan	62.42	Afghanistan	73.68	5.37	68.89	77.25	8.36
2014	73.46	83.59	Japan	62.54	Afghanistan	73.86	5.41	69.07	77.45	8.38
2015	73.69	83.79	Japan	62.66	Afghanistan	73.88	5.36	69.64	77.72	8.08

- **Steady Upward Trend:** The mean life expectancy in Asia improved from 69.13 (2000) to 73.69 (2015) — a +4.56 year increase. This is slightly below the improvement seen in Africa (+7.66 years), but Asia started from a higher baseline.
- **Highest Values – Japan Dominates:** Japan consistently had the highest life expectancy every year, starting at 81.08 (2000) and climbing to 83.79 (2015) — consistently above the global average and setting the regional benchmark.
- **Lowest Values – Afghanistan:** *Afghanistan* remained the lowest-performing country for nearly all years except 2008, when Myanmar briefly held the lowest spot.
 - In 2000, Afghanistan's life expectancy was 55.3
 - By 2015, it had increased to 62.66 While still the lowest in Asia, this is a +7.36 year gain, aligning with African progress.
- **Median Life Expectancy:** The median increased from 70.36 (2000) to 73.88 (2015), indicating that more than half of Asian countries reached above 73.88 years by 2015.
- **Slight Decrease in Dispersion:** The standard deviation decreased from 5.96 to 5.36, and the IQR (Interquartile Range) from 8.58 to 8.08, pointing to modest convergence in health outcomes across the region.
- **Percentile Shifts:**
 - 25th percentile improved from 64.89 to 69.64
 - 75th percentile rose from 73.47 to 77.72
 - This shift confirms broad-based regional improvements, particularly among countries in the lower half of the distribution.

Contextual Notes:

- Asia started from a higher life expectancy baseline than Africa and has maintained this lead.
- The most significant improvements occurred in countries with initially low life expectancies, reducing the region's overall disparity.

America

year	mean	max	max_countries	minimum	min_countries	median	std	25%	75%	iqr
2000	71.14	79.17	Canada	58.36	Haiti	71.32	4.5	68.88	74.04	5.17
2001	71.37	79.4	Canada	58.49	Haiti	71.5	4.49	69.31	74.33	5.02
2002	71.67	79.53	Canada	58.86	Haiti	71.94	4.5	69.7	74.61	4.91
2003	71.81	79.72	Canada	59.51	Haiti	72.35	4.46	69.62	74.84	5.22
2004	72.06	79.99	Canada	58.48	Haiti	72.48	4.54	70.07	75.2	5.13
2005	72.41	80.11	Canada	60.42	Haiti	72.99	4.34	70.27	75.48	5.21
2006	72.68	80.54	Canada	60.76	Haiti	73.17	4.24	70.52	75.6	5.08
2007	72.92	80.54	Canada	61.1	Haiti	73.22	4.11	70.88	75.74	4.85
2008	73.17	80.72	Canada	61.33	Haiti	73.38	4.07	71.26	75.93	4.67
2009	73.5	81.07	Canada	61.74	Haiti	73.47	3.99	71.61	76.14	4.53
2010	73.13	81.32	Canada	46.02	Haiti	73.71	5.94	71.95	76.09	4.14
2011	73.89	81.48	Canada	61.62	Haiti	74.13	3.99	72.28	76.38	4.1
2012	74.06	81.66	Canada	62.29	Haiti	74.15	3.92	72.53	76.64	4.11
2013	74.22	81.74	Canada	62.6	Haiti	73.96	3.89	72.72	76.76	4.05
2014	74.36	81.78	Canada	62.99	Haiti	74.47	3.83	72.84	77.0	4.17
2015	74.48	81.82	Canada	63.24	Haiti	74.5	3.78	72.74	77.13	4.39

- **Gradual Improvement Across the Board:** Life expectancy in the Americas improved steadily from 71.14 (2000) to 74.48 (2015) — a +3.34 year increase. While lower than Africa and Asia in raw gain, this region already had a relatively high starting point.
- **Highest Values** – Canada Consistently on Top: Canada held the highest life expectancy throughout the 16-year period:
 - From 79.17 (2000) to 81.82 (2015) This is on par with top Asian performers like Japan.
- **Lowest Values** – Haiti: Haiti consistently had the lowest life expectancy in the region (except in 2010, when its value anomalously dropped to 46.02—likely reflecting the catastrophic earthquake impact that year).
 - Life expectancy improved from 58.36 (2000) to 63.24 (2015) This +4.88 year gain is respectable but still leaves Haiti well below the regional average.
- **Median Life Expectancy:** The median rose from 71.32 to 74.50, suggesting a broad regional uplift and narrowing gap between nations.
- **Moderate Decrease in Dispersion:**
 - Standard deviation dropped from 4.5 to 3.78
 - Interquartile range (IQR) decreased from 5.17 to 4.39 This reflects a converging trend in life expectancy, with countries drawing closer together in longevity outcomes.
- **Percentile Improvements:**
 - 25th percentile: 68.88 → 72.74
 - 75th percentile: 74.04 → 77.13
 - These shifts affirm a general upward movement for all countries, regardless of their 2000 starting position.

Contextual Notes:

- The Americas already had a mature health system landscape in 2000, so gains were smaller in percentage terms.
- Haiti's 2010 drop is a clear outlier due to external disaster, but the country rebounded strongly in subsequent years.
- North American countries (Canada, USA) pull the upper end up, while Caribbean and parts of Central America influence the lower end.

Europe

year	mean	max	max_countries	minimum	min_countries	median	std	25%	75%	iqr
2000	74.86	79.78	Italy	65.48	Russian Federation	75.41	3.89	71.84	77.96	6.12
2001	75.2	80.69	Iceland	65.38	Russian Federation	75.76	3.98	72.24	78.36	6.12
2002	75.29	80.5	Iceland	65.13	Russian Federation	76.01	4.0	72.32	78.46	6.15
2003	75.48	80.96	Iceland	65.03	Russian Federation	76.86	3.98	72.37	78.52	6.15
2004	75.93	81.09	Switzerland	65.47	Russian Federation	77.21	4.04	72.67	79.11	6.44
2005	76.02	81.5	Iceland	65.53	Russian Federation	77.61	4.14	72.74	79.32	6.57
2006	76.32	81.49	Switzerland	66.73	Russian Federation	78.09	4.13	73.24	79.44	6.19
2007	76.54	81.74	Switzerland	67.59	Russian Federation	78.2	4.09	73.39	79.78	6.39
2008	76.89	81.99	Switzerland	67.95	Russian Federation	78.45	3.99	73.83	80.18	6.34
2009	77.18	82.04	Switzerland	68.68	Russian Federation	78.6	3.86	74.22	80.28	6.06
2010	77.49	82.25	Switzerland	68.84	Russian Federation	79.03	3.85	74.67	80.66	5.99
2011	77.9	82.7	Switzerland	69.19	Republic of Moldova	79.8	3.85	74.98	80.96	5.99
2012	78.03	82.92	Iceland	69.14	Republic of Moldova	80.05	3.75	74.98	80.92	5.94
2013	78.29	83.08	Spain	69.1	Republic of Moldova	80.3	3.73	75.48	81.3	5.81
2014	78.57	83.23	Spain	69.03	Republic of Moldova	80.7	3.87	75.56	81.6	6.04
2015	78.52	82.9	Switzerland	69.24	Republic of Moldova	80.64	3.71	75.48	81.5	6.01

- **High Starting Point and Steady Growth:** Europe began with a strong average life expectancy of 74.86 years in 2000, rising to 78.52 by 2015 — a +3.66 year increase, slightly more than in the Americas but less than Asia and Africa.
- **Top Performers** – Western Europe Leads: Countries like Switzerland, Iceland, and Spain consistently topped the charts:
 - Switzerland held the highest life expectancy in 9 of the 16 years, peaking at 83.23 (2014).
 - Iceland led early on, with values exceeding 80 by 2001.
- **Lowest Values** – East European Struggles:
 - From 2000 to 2010, the Russian Federation had the lowest life expectancy, hovering around 65 years.
 - Republic of Moldova replaced it from 2011 to 2015, ending at 69.24 years in 2015.
 - These figures are more than 14 years lower than the highest values in Western Europe.
- **Median Life Expectancy:** Rose from 75.41 in 2000 to 80.64 in 2015, reflecting widespread longevity gains across the region — especially strong in Western and Northern Europe.
- **Distribution and Dispersion:**
 - Standard deviation fluctuated slightly but decreased overall from 3.89 → 3.71, indicating a modest convergence among countries.
 - Interquartile range (IQR) remained relatively stable, with a slight dip: 6.12 → 6.01
- **Percentile Shift:**
 - 25th percentile: 71.84 → 75.48
 - 75th percentile: 77.96 → 81.50
 - The upward shift across percentiles shows that both lower- and upper-quartile countries made significant progress.

Contextual Notes:

Europe exhibited one of the most consistent and stable progressions in life expectancy, benefiting from mature healthcare systems. The gap between East and West remained evident, but the difference narrowed slightly over time. The post-Soviet states, especially Moldova and Russia, faced persistent challenges despite modest improvements.

Oceania

year	mean	max	max_countries	minimum	min_countries	median	std	25%	75%	iqr
2000	69.6	79.23	Australia	61.72	Papua New Guinea	68.97	5.31	66.54	70.5	3.97
2001	69.81	79.63	Australia	61.77	Papua New Guinea	69.09	5.36	66.6	70.88	4.29
2002	69.79	79.94	Australia	61.7	Papua New Guinea	69.26	5.47	66.19	70.82	4.63
2003	70.1	80.24	Australia	61.8	Papua New Guinea	69.39	5.52	66.54	71.32	4.77
2004	70.25	80.49	Australia	61.76	Papua New Guinea	69.49	5.62	66.76	71.48	4.72
2005	70.38	80.84	Australia	61.8	Papua New Guinea	69.58	5.71	66.96	71.53	4.58
2006	70.53	81.04	Australia	61.92	Papua New Guinea	69.68	5.75	67.17	71.79	4.62
2007	70.59	81.29	Australia	62.03	Papua New Guinea	69.8	5.81	67.12	71.84	4.72
2008	70.76	81.4	Australia	62.57	Papua New Guinea	69.88	5.76	67.31	71.9	4.6
2009	70.63	81.54	Australia	62.79	Papua New Guinea	69.83	5.78	67.44	70.85	3.41
2010	71.01	81.7	Australia	63.04	Papua New Guinea	69.97	5.78	67.68	72.0	4.31
2011	71.17	81.9	Australia	63.53	Papua New Guinea	69.96	5.76	67.87	72.03	4.16
2012	71.27	82.05	Australia	63.73	Papua New Guinea	69.97	5.78	67.93	71.98	4.05
2013	71.36	82.15	Australia	63.96	Papua New Guinea	70.02	5.75	67.87	72.17	4.3
2014	71.44	82.3	Australia	64.26	Papua New Guinea	70.0	5.73	67.84	72.23	4.38
2015	71.52	82.4	Australia	64.7	Papua New Guinea	70.1	5.73	67.51	72.24	4.73

- **Gradual Increase with Wide Gaps:** Oceania's regional average life expectancy increased steadily from 69.60 years in 2000 to 71.52 in 2015, a +1.92 gain over period — the smallest increase among all global regions.
- **Top Performer – Australia's Dominance:**
 - Australia consistently had the highest life expectancy across all 16 years.
 - It rose from 79.23 in 2000 to 82.40 in 2015, demonstrating a strong and steady trend of improvement.
- **Lowest Values – Persistent Struggles in Papua New Guinea:**
 - Papua New Guinea had the lowest life expectancy for every year
 - Life expectancy in Papua New Guinea improved from 61.72 to 64.70, but it remained over 17 years behind Australia by 2015.
- **Median Life Expectancy:** Increased moderately from 68.97 in 2000 to 70.10 in 2015, indicating overall upward mobility, though the gains were modest and uneven.
- **Distribution and Dispersion:**
 - Standard deviation remained high and relatively constant: 5.31 → 5.73, reflecting a wide disparity between nations.
 - The gap between top and bottom countries persisted across the time span.
- **Percentile Ranges:**
 - 25th percentile: 66.54 → 67.51
 - 75th percentile: 70.50 → 72.24
 - IQR (Interquartile Range) hovered around 4 to 4.7 years, showing moderate inequality that did not shrink over time.

Contextual Notes:

- Oceania shows sharp duality:
 - High-income countries like Australia enjoy advanced healthcare and long life spans.
 - In contrast, developing island nations — notably Papua New Guinea — face infrastructure and access challenges.
- The consistently high dispersion metrics highlight this persistent health gap.

3.1.2 GDP per Capita

Global Data:

year	mean	max	max_countries	minimum	min_countries	median	std	25%	75%	iqr
2000	6432.78	48659.6	Luxembourg	122.96	Ethiopia	1673.36	9895.37	611.47	6526.71	5915.24
2001	6342.96	48440.14	Luxembourg	119.26	Ethiopia	1800.4	9681.83	596.62	6529.2	5932.58
2002	6708.98	53005.73	Luxembourg	110.46	Ethiopia	1989.14	10365.15	658.98	6610.02	5951.03
2003	7824.77	65689.32	Luxembourg	114.37	Burundi	2259.52	12224.57	704.9	8272.54	7567.64
2004	9007.8	76544.92	Luxembourg	128.54	Burundi	2600.37	14030.32	807.84	9739.93	8932.09
2005	9903.23	80988.14	Luxembourg	151.19	Burundi	3106.58	15177.77	980.34	10834.92	9854.58
2006	10802.97	90788.8	Luxembourg	166.28	Burundi	3343.79	16326.49	1166.53	12072.4	10905.87
2007	12311.73	107475.32	Luxembourg	170.71	Burundi	3934.69	18453.09	1317.42	14027.22	12709.8
2008	13629.04	120422.14	Luxembourg	194.71	Burundi	4504.51	20067.95	1465.33	17018.96	15553.63
2009	11958.56	109419.75	Luxembourg	204.54	Burundi	4270.86	17438.69	1366.78	14040.76	12673.99
2010	12747.37	110885.99	Luxembourg	222.66	Burundi	4835.79	18282.4	1512.32	14180.24	12667.93
2011	14241.82	119025.06	Luxembourg	236.45	Burundi	5614.39	20571.54	1673.25	14874.8	13201.54
2012	14154.42	112584.68	Luxembourg	238.21	Burundi	6015.95	20207.19	1738.04	15926.34	14188.3
2013	14516.97	120000.14	Luxembourg	241.55	Burundi	6141.98	20671.21	1871.68	16446.79	14575.11
2014	14592.0	123678.7	Luxembourg	257.82	Burundi	6164.64	20643.45	1826.94	16677.97	14851.03
2015	12928.92	105462.01	Luxembourg	289.36	Burundi	5588.98	17920.52	1676.38	15659.32	13982.95

- **Consistent Upward Trend Until 2014, Followed by a Decline:** Global average GDP per Capita rose steadily from 6,432.78 in 2000 to a high of 14,592.00 in 2014 — more than doubling over the period. However, it declined in 2015 to \$12,928.92, likely reflecting global economic shifts or downturns.
- **Top Performer** – Luxembourg Dominates:
 - Luxembourg had the highest GDP per Capita in all 16 years, starting at 48,659.60 in 2000 and peaking at 123,678.70 in 2014.
 - Its consistent lead underscores its strong finance-driven economy and small population.
- **Lowest GDP per Capita** – Persistent Struggles in Sub-Saharan Africa:
 - Ethiopia recorded the lowest values in 2000 and 2001 (as low as \$119.26).
 - From 2003 to 2015, Burundi consistently had the lowest GDP per Capita, reaching just \$289.36 by 2015 — over 400 times lower than Luxembourg in the same year.
- **Median GDP per Capita:** Rose from 1,673.36 in 2000 to 6,164.64 in 2014, falling slightly to \$5,588.98 in 2015. This steady rise shows that typical countries also experienced income growth, not just the top performers.
- **Distribution and Dispersion:**
 - **Standard deviation** widened from 9,895.37 → 20,643.45 (2014), indicating increasing global inequality in income per person.
 - It slightly narrowed in 2015 to \$17,920.52, possibly due to declining GDP among top economies.
- **Interquartile Range (IQR):**
 - Grew substantially from 5,915.24 in 2000 to 14,851.03 in 2014, narrowing in 2015 to \$13,982.95.
 - This suggests a widening income gap among middle-income countries over time, followed by slight convergence.
- **Percentile Shift:**
 - **25th percentile:** 611.47 → 1,826.94
 - **75th percentile:** 6,526.71 → 16,677.97
 - This upward shift across both quartiles reveals that income growth occurred across many nations, not just among wealthy ones.

Contextual Notes

The dataset reflects strong global economic growth from 2000 to 2014, but also an increasing income disparity between nations. The dip in 2015 may reflect effects of global financial adjustments, commodity price shocks, or currency fluctuations. Persistent low GDP per Capita in Sub-Saharan countries highlights ongoing development challenges, while Luxembourg's dominance reflects the impact of high-income microstates on global statistics.

Africa

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	1149.94	8063.66	Seychelles	122.96	Ethiopia	527.04	1651.87	289.57	1140.02	850.45
2001	1118.55	8153.3	Seychelles	119.26	Ethiopia	498.42	1571.84	300.52	1150.04	849.52
2002	1112.41	8864.17	Seychelles	110.46	Ethiopia	529.32	1503.12	312.77	1113.64	800.86
2003	1316.89	9070.28	Seychelles	114.37	Burundi	593.98	1713.2	358.27	1144.03	785.76
2004	1581.19	10827.67	Seychelles	128.54	Burundi	731.86	2099.11	405.98	1302.75	896.76
2005	1850.53	11802.11	Seychelles	151.19	Burundi	824.93	2549.48	448.13	1878.19	1430.06
2006	2060.53	12782.99	Seychelles	166.28	Burundi	906.0	2846.94	449.74	2194.29	1744.55
2007	2319.69	13776.9	Equatorial Guinea	170.71	Burundi	962.34	3158.59	510.22	2618.54	2108.32
2008	2691.22	19849.72	Equatorial Guinea	194.71	Burundi	1115.3	3821.93	599.05	3097.29	2498.24
2009	2328.34	14398.77	Equatorial Guinea	204.54	Burundi	1094.13	2906.66	640.15	2909.59	2269.44
2010	2597.29	14905.51	Equatorial Guinea	222.66	Burundi	1166.32	3264.93	672.5	3041.58	2369.08
2011	2852.38	18659.42	Equatorial Guinea	236.45	Burundi	1294.3	3642.85	721.24	3386.91	2665.66
2012	3002.71	18756.43	Equatorial Guinea	238.21	Burundi	1289.99	3977.07	706.6	3478.6	2772.0
2013	3023.63	17644.59	Equatorial Guinea	241.55	Burundi	1384.88	3807.83	727.65	3634.15	2906.5
2014	2983.76	16804.92	Equatorial Guinea	257.82	Burundi	1450.63	3646.81	735.38	3575.5	2840.12
2015	2530.5	15333.11	Seychelles	289.36	Burundi	1315.43	2980.87	705.62	3161.62	2456.0

- **Steady Growth Followed by Decline:** Average GDP per Capita in Africa increased from 1, 149.94 in 2000 to a peak of 3,002.71 in 2012, then slightly declined to \$2,530.50 in 2015. This reflects solid economic growth across the continent, tempered by external shocks in the later years.
- **Top Performers** – Seychelles and Equatorial Guinea:
 - Seychelles led in the early 2000s and re-emerged as the top in 2015, with GDP per Capita peaking at \$8,864.17 in 2002.
 - Equatorial Guinea dominated from 2007 to 2014, reaching a high of \$19,849.72 in 2008 due to oil revenue booms.
- **Lowest Values** – Burundi and Ethiopia:
 - Ethiopia had the lowest GDP per Capita in 2000–2002, bottoming at \$110.46 in 2002.
 - From 2003 to 2015, Burundi consistently ranked lowest, with only \$289.36 in 2015 — over 50 times less than the region's top performers.
- **Median GDP per Capita:** Rose from 527.04 in 2000 to 1,450.63 in 2014, before dropping to \$1,315.43 in 2015. This consistent rise suggests broad-based economic improvement, although progress slowed in the final year.
- **Distribution and Dispersion:**
 - **Standard deviation** increased from 1, 651.87 → 3,977.07 (2012), then narrowed to \$2,980.87 in 2015, indicating growing inequality during peak growth years, followed by some convergence.
 - **Interquartile range (IQR)** rose from 850.45 in 2000 to 2,906.50 in 2013, dropping slightly to \$2,456.00 in 2015 — reflecting widening and then stabilizing disparities among countries.

- **Percentile Shift:**
 - **25th percentile:** 289.57 → 705.62
 - **75th percentile:** 1, 140.02 → 3,161.62
 - These shifts demonstrate that both lower- and upper-income African countries saw significant gains during the period, though gaps persisted.

Contextual Notes:

- Africa experienced significant economic expansion, especially during the global commodity boom years (2004–2012).
- Despite overall growth, deep inequalities remained, with a handful of resource-rich countries far outpacing others.
- Burundi and Ethiopia highlight ongoing poverty challenges, while Equatorial Guinea and Seychelles represent exceptional outliers.

The decline in 2015 may reflect external global economic pressures, including falling oil prices and reduced foreign investment.

Asia

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	6157.41	39169.36	Japan	137.18	Tajikistan	1161.24	9537.13	558.23	8321.39	7763.16
2001	5798.11	34406.18	Japan	140.78	Myanmar	1209.77	8746.67	534.65	8191.51	7656.86
2002	5934.89	32820.79	Japan	145.82	Myanmar	1204.23	8766.19	559.83	8380.04	7820.21
2003	6592.25	35387.04	Japan	199.64	Afghanistan	1256.03	9736.92	633.95	8896.9	8262.95
2004	7623.02	40792.28	Qatar	221.83	Afghanistan	1418.81	11322.84	774.79	10030.44	9255.65
2005	8771.55	52468.45	Qatar	251.17	Myanmar	1744.51	13065.05	832.75	12357.7	11524.95
2006	9771.7	59978.86	Qatar	274.02	Afghanistan	2253.79	14409.82	909.03	14533.73	13624.7
2007	10731.43	64706.99	Qatar	376.32	Afghanistan	3064.28	15247.79	1022.73	15756.19	14733.46
2008	12344.32	79811.6	Qatar	382.53	Afghanistan	3908.95	17538.35	1158.1	18944.86	17786.76
2009	10450.48	60733.98	Qatar	453.39	Afghanistan	3832.23	13996.66	1225.85	15064.63	13838.78
2010	11992.17	73021.31	Qatar	562.5	Afghanistan	4430.43	16005.03	1742.35	17958.94	16216.59
2011	14035.38	92993.0	Qatar	608.74	Afghanistan	5453.16	19303.54	2051.13	22441.57	20390.44
2012	14452.71	98041.36	Qatar	653.42	Afghanistan	6049.47	19944.62	2190.23	24069.2	21878.97
2013	14353.15	97630.83	Qatar	638.73	Afghanistan	6171.3	19560.2	2367.5	23563.94	21196.44
2014	14132.4	93126.15	Qatar	626.51	Afghanistan	5822.38	18909.55	2558.78	23121.21	20562.43
2015	12150.58	66984.91	Qatar	566.88	Afghanistan	4990.94	15303.2	2595.23	18777.43	16182.2

- **Steady Growth With Regional Peaks:** Average GDP per Capita in Asia increased from 6,157.41 in 2000 to a peak of 14,452.71 in 2012, before declining to \$12,150.58 in 2015. The growth pattern reflects overall economic development, with fluctuations influenced by regional economic factors and global economic pressures.
- **Top Performers – Qatar and Japan:**
 - Qatar was the dominant performer throughout the period, consistently ranking highest in GDP per Capita, peaking at \$98,041.36 in 2012.
 - Japan, though a top performer, showed a steady decline after 2007, reaching a high of 39,169.36 in 2000, before stabilizing in the 30,000 range post-2008.

- **Lowest Values** – Afghanistan and Myanmar:
 - Afghanistan had the lowest GDP per Capita across the years, bottoming out at \$137.18 in 2000 and remaining near the lowest values throughout.
 - Myanmar's GDP per Capita was also consistently low, with values ranging from 140.78 in 2001 to a peak of 608.74 in 2011, still far below regional averages.
- **Median GDP per Capita:** Rose from 1,161.24 in 2000 to 6,049.47 in 2012, before dropping to \$4,990.94 in 2015. The significant increase until 2012 points to broad improvements in middle-income countries, with a slowdown in growth post-2012.
- **Distribution and Dispersion:**
 - **Standard deviation** increased from 9,537.13 → 19,944.62 (2012), then narrowed to \$15,303.20 in 2015, indicating rising inequality followed by some degree of convergence toward the later years.
 - **Interquartile range (IQR)** rose from 7,763.16 in 2000 to 21,196.44 in 2012, before narrowing to \$16,182.20 in 2015 — reflecting growing disparities in economic development during peak growth years, followed by stabilization.
- **Percentile Shift:**
 - 25th percentile: 558.23 → 2,595.23
 - 75th percentile: 8,321.39 → 23,563.94
 - These shifts demonstrate that both lower- and upper-income Asian countries saw significant gains during the period, with widening income gaps until 2012, followed by some convergence post-2012.

Contextual Notes:

- Asia experienced significant economic growth over the period, with countries like Qatar, Japan, and emerging economies such as China and India contributing significantly to the rise in GDP per Capita.
- However, disparities remained, with countries like Afghanistan and Myanmar struggling with low levels of economic development.

The decline post-2012 could be attributed to external factors such as the global economic slowdown, oil price volatility, and political instability in some regions.

America

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	6368.22	36329.97	United States of America	815.0	Haiti	3760.08	7958.67	1834.83	6526.71	4691.88
2001	6338.43	37133.62	United States of America	743.91	Haiti	3841.6	8048.71	1756.06	6461.87	4705.91
2002	6256.67	37997.74	United States of America	716.48	Haiti	3683.89	8317.14	2036.91	5277.45	3240.54
2003	6586.55	39490.3	United States of America	575.56	Haiti	3557.2	8764.9	2161.81	5745.56	3583.75
2004	7253.43	41724.64	United States of America	679.28	Haiti	4130.1	9328.34	2333.82	6339.85	4006.14
2005	7929.84	44123.4	United States of America	771.53	Haiti	4773.27	10082.28	2568.28	7213.32	4645.04
2006	8780.45	46301.99	United States of America	824.36	Haiti	5385.56	10685.7	3130.7	8518.59	5387.89
2007	9650.31	48050.23	United States of America	979.6	Haiti	6055.09	11309.2	3535.04	9077.12	5542.08
2008	10387.86	48570.06	United States of America	1089.58	Haiti	6641.5	11490.81	4201.16	9959.01	5757.84
2009	9616.42	47194.95	United States of America	1191.8	Haiti	6737.86	10484.25	4120.6	9175.32	5054.72
2010	10543.13	48650.66	United States of America	1204.86	Haiti	6760.97	11235.69	4657.8	11880.94	7223.14
2011	11414.33	52223.86	Canada	1306.85	Haiti	7392.94	11827.41	5221.77	13919.16	8697.38
2012	11844.15	52670.34	Canada	1356.17	Haiti	8096.8	12076.85	5539.25	14109.75	8570.5
2013	12168.15	53409.75	United States of America	1452.31	Haiti	8263.64	12186.5	5720.86	14028.66	8307.8
2014	12298.39	55304.32	United States of America	1454.65	Haiti	8167.47	12239.59	5763.46	13756.2	7992.74
2015	11950.56	57040.21	United States of America	1495.71	Haiti	8379.62	11806.01	5764.92	13729.31	7964.4

- **Steady Climb with Minor Setbacks:** The Americas saw a general upward trend in average GDP per capita, rising from 6,368.22 in 2000 to a peak of 12,298.39 in 2014, before a slight dip to \$11,950.56 in 2015. The region's growth was largely driven by high-income countries, with some fluctuation due to global economic events such as the 2008 financial crisis and falling commodity prices.
- **Top Performers** – United States and Canada:
 - The United States led in most years, starting at 36,329.97 in 2000 and rising to 57,040.21 in 2015.
 - Canada briefly surpassed the U.S. in 2011 and 2012, with a high of \$52,670.34 in 2012, reflecting its stable economy and resource-driven wealth during that period.
- **Lowest Values – Haiti:**
 - **Haiti** consistently had the lowest GDP per capita across all years, starting at 815.00 in 2000 and ending at 1,405.71 in 2015. Despite slight improvements, its figures remained significantly lower than the regional average, highlighting long-standing economic challenges.
- **Median GDP per Capita:** The median rose from 3,760.08 in 2000 to 8,379.62 in 2015, indicating that most countries experienced solid gains, although not as extreme as top performers. The steady rise supports a moderate but widespread economic advancement across the region.
- **Distribution and Dispersion:**
 - **Standard deviation** widened from 7,958.67 in 2000 to a high of 12,239.59 in 2014, slightly falling to \$11,806.01 in 2015. This reflects increasing inequality, with richer countries accelerating faster than their poorer counterparts.
 - **Interquartile range (IQR)** expanded from 4,691.88 in 2000 to 8,697.38 in 2011, and stabilized around \$7,964.40 in 2015, showing a growing and then steady income gap between the lower and upper-middle income countries.
- **Percentile Shift:**
 - **25th percentile:** Grew from 1,834.83 in 2000 to 5,764.92 in 2015
 - **75th percentile:** Rose from 6,526.71 to 13,729.31 over the same period
 - This widening gap confirms that while economic growth was region-wide, the benefits were skewed toward upper-tier economies.

Contextual Notes:

- The Americas region showcases sharp contrasts: high-income countries like the U.S. and Canada pulling the average upward, while lower-income countries like Haiti lagged far behind.
- The 2008 global financial crisis caused a temporary dip, but the region rebounded by 2010. Resource-rich nations in South America benefited from commodity booms, though gains slowed in later years due to price drops and political instability.
- Haiti remains a major outlier, with persistent poverty and vulnerability to natural disasters and political turmoil hampering growth despite international aid and development efforts.

Europe

```
display(  
  Image(filename="../Visualizations/Dataframes/GDP_per_Capita/Descriptive_stats-GDP_per_Capi  
)
```

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	14021.78	48659.6	Luxembourg	440.54	Republic of Moldova	10201.3	13208.6	1816.74	24485.75	22669.01
2001	14161.5	48440.14	Luxembourg	507.4	Republic of Moldova	10402.23	13038.96	2004.98	24763.28	22758.3
2002	15622.24	53005.73	Luxembourg	570.8	Republic of Moldova	11289.89	14204.88	2330.69	26762.67	24431.98
2003	19087.45	65689.32	Luxembourg	682.32	Republic of Moldova	13669.5	17104.4	2990.27	32610.86	29620.59
2004	22109.08	76544.92	Luxembourg	897.18	Republic of Moldova	15197.06	19575.97	3802.58	37330.7	33528.12
2005	23646.82	80988.14	Luxembourg	1034.39	Republic of Moldova	15888.17	20837.57	4970.73	38736.16	33765.43
2006	25527.19	90788.8	Luxembourg	1183.02	Republic of Moldova	16723.88	22290.87	6338.86	40945.96	34607.1
2007	29862.62	107475.32	Luxembourg	1531.22	Republic of Moldova	19485.87	25462.56	8730.79	47695.86	38965.07
2008	32616.87	120422.14	Luxembourg	2110.55	Republic of Moldova	22804.58	27155.5	11035.25	50111.69	39076.44
2009	28668.53	109419.75	Luxembourg	1898.44	Republic of Moldova	21083.28	24187.22	8555.44	43249.02	34693.59
2010	28914.72	110885.99	Luxembourg	2436.8	Republic of Moldova	21798.91	24900.77	9536.4	43711.01	34174.61
2011	31782.12	119025.06	Luxembourg	2941.36	Republic of Moldova	23155.1	27554.36	11449.56	47562.58	36113.02
2012	30157.16	112584.68	Luxembourg	3044.81	Republic of Moldova	20563.71	26467.06	10957.78	45333.06	34375.27
2013	31534.8	120000.14	Luxembourg	3321.04	Republic of Moldova	21653.2	27468.46	11527.78	48281.46	36753.69
2014	32158.49	123678.7	Luxembourg	3104.65	Ukraine	21616.71	27730.93	12033.6	49175.56	37141.96
2015	28161.76	105462.01	Luxembourg	2124.66	Ukraine	18083.88	24297.85	9127.33	43498.87	34371.54

- **Strong Growth with Deep Inequality:** Europe displayed a robust upward trend in GDP per capita, with the regional average rising from 14,021.78 in 2000 to a peak of 32,158.49 in 2014. The average dipped to \$28,161.76 in 2015, likely influenced by external economic shocks and regional instabilities such as the Eurozone crisis. Despite the setbacks, the continent maintained a high overall economic standing throughout the period.
- **Top Performer** – Luxembourg:
 - Luxembourg overwhelmingly led the region every year, starting at 48,659.60 in 2000 and peaking at 123,678.70 in 2014.
 - It remained the highest performer through 2015, finishing at \$105,462.01, reflecting its strong financial sector, low population, and high productivity.
- **Lowest Values** – Republic of Moldova and Ukraine:
 - From 2000 to 2013, Republic of Moldova recorded the lowest GDP per capita in Europe, starting at just 440.54 in 2000 and gradually rising to 3,321.04 by 2013.
 - In 2014 and 2015, Ukraine took over as the lowest performer, with GDP per capita of 3,104.65 and 2,124.66 respectively, reflecting political unrest and economic decline during that time.

- **Median GDP per Capita:**
 - The median followed an upward trend from 10,201.30 *in 2000 to* 21,616.71 in 2014, before dropping to \$18,083.88 in 2015.
 - This trend suggests that while most countries experienced growth, the impact of regional downturns was felt more strongly by mid-income nations.
- **Distribution and Dispersion:**
 - Standard deviation increased from 13,208.60 *in 2000 to* 27,730.93 in 2014, dipping slightly to \$24,297.85 in 2015. This reflects the extreme spread between high- and low-income countries within the region.
 - **Interquartile range (IQR)** also grew from 22,669.01 *in 2000 to* 39,076.44 in 2008, remaining above \$34,000 through 2015.
 - The widening IQR shows that income disparities between the 25th and 75th percentile countries became more pronounced.
- **Percentile Shift:**
 - **25th percentile:** Rose from 1,816.74 *in 2000 to* 9,127.33 in 2015
 - **75th percentile:** Climbed from 24,485.75 *to* 43,498.87 over the same period
 - These changes signal that even lower-income countries saw significant gains, though not nearly as rapidly as the top performers.

Contextual Notes:

- Europe's wealth is largely driven by countries in Western and Northern Europe, such as Luxembourg, Switzerland, and Norway. These nations consistently elevate regional averages, even as Eastern and Southeastern Europe lag behind.
- The 2008 financial crisis had a visible impact, with mean and median values dipping around 2009, but the region showed strong recovery by 2011. However, the downturn in 2015 suggests lasting effects of internal fiscal strains and geopolitical tensions (e.g., Russia-Ukraine conflict).
- While countries like Moldova and Ukraine remain outliers on the low end, even they experienced gradual growth, albeit slower and more volatile. The economic divide between Western and Eastern Europe continues to shape the region's overall statistical profile.

Oceania

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	4689.28	21870.42	Australia	639.28	Papua New Guinea	1706.36	6804.17	1084.01	2070.12	986.11
2001	4438.49	19695.73	Australia	540.67	Papua New Guinea	1595.09	6329.88	1024.66	2105.43	1080.77
2002	4815.73	20301.84	Australia	509.03	Papua New Guinea	1635.28	6949.16	933.38	2163.39	1230.01
2003	5824.45	23718.13	Australia	580.6	Papua New Guinea	1857.94	8526.04	1142.7	2559.0	1416.3
2004	7032.09	30836.73	Australia	624.04	Papua New Guinea	2161.91	10639.67	1243.53	2892.46	1648.92
2005	7776.81	34479.77	Australia	748.74	Papua New Guinea	2367.22	11789.51	1323.06	3186.8	1863.74
2006	8001.27	36595.71	Australia	1121.08	Kiribati	2469.22	12040.47	1430.94	3300.69	1869.75
2007	9185.16	41051.61	Australia	1353.62	Kiribati	2571.89	13942.9	1599.71	3597.6	1997.89
2008	10094.56	49701.28	Australia	1411.91	Kiribati	2873.08	15761.1	1831.47	3782.7	1951.24
2009	8987.75	42816.57	Australia	1324.44	Kiribati	2756.94	13674.36	1802.34	3234.38	1432.04
2010	10676.62	52147.02	Australia	1528.9	Kiribati	3088.06	16652.83	2092.57	3488.09	1395.52
2011	12475.98	62609.66	Australia	1783.77	Kiribati	3427.19	19782.72	2449.23	4083.58	1634.34
2012	13321.88	68078.04	Australia	1855.27	Kiribati	3462.95	21315.9	2716.4	4373.88	1657.47
2013	13632.62	68198.42	Australia	1780.66	Kiribati	3455.3	21738.84	2653.24	4492.27	1839.03
2014	13302.37	62558.24	Australia	1740.36	Kiribati	3438.59	20553.07	2771.99	5010.15	2238.16
2015	12048.34	56758.87	Australia	1641.48	Kiribati	3469.9	18306.33	2537.52	4858.38	2320.85

- **Uneven Growth with Heavy Skew Toward Australia:** Oceania's average GDP per capita showed an upward trajectory from 4,689.28 in 2000 to a peak of 13,632.62 in 2013, before a decline to \$12,048.34 in 2015. The rise reflects overall regional economic progress, though the average is heavily influenced by Australia's high figures. Smaller Pacific Island nations contributed less to the regional mean but saw steady improvements in their own metrics.
- **Top Performer – Australia:**
 - Australia led consistently throughout the period, starting at 21,870.42 in 2000 and peaking at 68,198.42 in 2013.
 - It remained the dominant economy through 2015, ending at \$56,758.87, significantly outpacing every other nation in the region due to its diversified economy, strong exports, and stable governance.
- **Lowest Values – Papua New Guinea and Kiribati:**
 - Papua New Guinea had the lowest GDP per capita from 2000 to 2005, beginning at \$639.28 in 2000.
 - From 2006 onward, Kiribati recorded the lowest values, reaching \$1,641.48 in 2015.
 - Despite small gains, these countries remained far behind the regional average, reflecting limited industrialization and economic dependence on aid and subsistence livelihoods.
- **Median GDP per Capita:**
 - The median GDP per capita rose from 1,706.36 in 2000 to 3,469.90 in 2015.
 - This steady increase suggests gradual economic improvements among the smaller island nations, even though they lagged far behind the Australian benchmark.
- **Distribution and Dispersion:**
 - **Standard deviation** widened from 6,804.17 in 2000 to 21,738.84 in 2013, before narrowing to \$18,306.33 in 2015. This reflects the extreme inequality driven primarily by Australia's outsized economy.
 - **Interquartile range (IQR)** expanded from 986.11 in 2000 to 2,320.85 in 2015, peaking at \$2,238.16 in 2014. The increase in IQR reveals a modest widening gap between lower- and upper-middle-income countries in the region.

- **Percentile Shift:**
 - **25th percentile:** Rose from 1,084.01 *in 2000 to* 2,537.52 in 2015
 - **75th percentile:** Increased from 2,070.12 *to* 4,858.38 during the same period
 - This indicates an upward shift across the board, with all countries making progress, albeit at widely varying rates.

Contextual Notes:

- Oceania presents a stark economic contrast between Australia and the smaller island nations. Australia's large GDP per capita consistently skews the regional average upward, overshadowing the modest gains made by others.
- The 2008 global financial crisis had a muted impact on Oceania, with Australia weathering it relatively well due to strong trade ties with Asia and a resilient commodity sector.
- Countries like Kiribati and Papua New Guinea remain economically vulnerable, with challenges stemming from geographic isolation, limited industrial bases, and reliance on subsistence economies. Nevertheless, even these nations saw gradual improvements in GDP per capita, highlighting slow but positive regional development.

3.2 Summary Statistics for Exploratory Analysis

3.2.1 GDP-Based Country Grouping

To investigate disparities in life expectancy outcomes based on economic standing, we created two distinct groups using GDP per capita as the metric for classification:

1. Top 20 GDP Countries:

This group includes all countries that ranked within the top 20 for GDP per capita at least once between 2000 and 2015.

Code snippet:

```
top_20_gdp_countries_list = life_expectancy.groupby('year').apply(lambda x:
x.nlargest(20, 'gdp_per_capita'))['country'].unique()
le_top_20_gdp =
life_expectancy[life_expectancy['country'].isin(top_20_gdp_countries_list)].reset_index
```

2. Bottom 20 GDP Countries:

This group captures countries that appeared in the bottom 20 for GDP per capita across any year in the same range.

Code snippet:

```
le_bottom_20_gdp_countries_list = life_expectancy.groupby('year').apply(lambda x:
x.nsmallest(20, 'gdp_per_capita'))['country'].unique()
le_bottom_20_gdp =
life_expectancy[life_expectancy['country'].isin(le_bottom_20_gdp_countries_list)].reset
```

Purpose of the Grouping:

This categorization enables us to:

- Compare economic disparity by isolating top- and bottom-performing economies over time.
- Assess how life expectancy correlates with sustained high or low GDP per capita.
- Control for volatility by focusing only on countries that consistently appeared in extreme GDP positions.

Summary Statistics Overview:

To describe the economic conditions of these two groups, we calculated summary statistics for GDP per capita across the years 2000 to 2015. These include:

- Mean GDP per capita
- Median, Minimum, and Maximum
- 25th and 75th percentiles
- Interquartile Range (IQR)
- Standard Deviation (std)
- Countries with Highest and Lowest GDP per year

This allows us to understand not only the central tendencies of each group but also the spread and consistency of their economic performance over time.

Top 20 GDP Countries – Summary Statistics (2000–2015):

year	mean	max	max_country	minimum	median	std	25%	75%	iqr
2000	77.66	81.08	Japan	44.52	77.94	2.01	76.66	79.14	2.48
2001	77.96	81.42	Japan	41.96	78.22	2.07	76.95	79.34	2.39
2002	78.1	81.69	Japan	44.56	78.26	2.06	77.14	79.46	2.32
2003	78.32	81.76	Japan	43.39	78.47	2.01	77.29	79.64	2.35
2004	78.74	82.03	Japan	44.5	79.11	2.1	77.75	80.12	2.36
2005	78.94	81.96	Japan	44.77	79.34	2.09	78.09	80.15	2.06
2006	79.18	82.32	Japan	45.36	79.54	2.13	78.36	80.7	2.34
2007	79.39	82.51	Japan	45.61	79.94	2.18	78.46	80.81	2.34
2008	79.61	82.59	Japan	46.72	80.3	2.21	78.73	81.02	2.29
2009	79.83	82.93	Japan	48.06	80.44	2.24	78.88	81.32	2.44
2010	80.06	82.84	Japan	49.26	80.66	2.22	79.29	81.52	2.23
2011	80.36	82.7	Switzerland	49.95	80.98	2.28	79.96	81.78	1.83
2012	80.5	83.1	Japan	50.54	81.02	2.29	80.13	81.9	1.77
2013	80.65	83.33	Japan	50.78	81.22	2.28	80.35	82.04	1.69
2014	80.98	83.59	Japan	50.57	81.47	2.34	80.8	82.29	1.49
2015	80.91	83.79	Japan	51.59	81.5	2.34	80.66	82.32	1.66

The top 20 GDP group maintained a consistently high life expectancy, with the mean increasing from 77.66 years in 2000 to 80.91 years in 2015. This upward trend illustrates improved healthcare systems, higher living standards, and effective policy implementations in wealthier nations.

- **The maximum life expectancy** across this group consistently exceeded 81 years, peaking at 83.79 years in 2015, with Japan dominating as the country with the highest life expectancy for most years. Notably, Switzerland led briefly in 2011.
- The **minimum life expectancy**, although lower, gradually rose from 44.52 years (2000) to 51.59 years (2015), suggesting that even in wealthier nations, disparities still existed — possibly due to countries entering the top 20 GDP group despite having marginalized populations or uneven healthcare access.
- **Standard deviation** remained low (~2.0–2.3), indicating low variability and a tight clustering of life expectancy around the mean — a hallmark of stability in high-income countries.

- **The interquartile range (IQR)** declined over time (from 2.48 in 2000 to 1.66 in 2015), signaling an increase in equality of life expectancy among the richest nations.

These statistics emphasize the advantage that higher income levels provide in sustaining longevity, especially through uniform access to healthcare and education.

Bottom 20 GDP Countries – Summary Statistics (2000–2015)

year	mean	max	minimum	min_country	median	std	25%	75%	iqr
2000	53.39	70.76	44.52	Malawi	51.57	6.57	48.58	58.14	9.55
2001	53.9	71.64	41.96	Zimbabwe	52.39	6.72	49.22	58.46	9.23
2002	54.59	71.94	44.56	Zimbabwe	52.8	6.51	50.57	58.82	8.25
2003	55.19	72.41	43.39	Zimbabwe	53.2	6.58	51.0	59.4	8.39
2004	55.94	72.47	44.5	Zimbabwe	54.8	6.43	52.04	59.91	7.86
2005	56.56	72.77	44.77	Zimbabwe	55.3	6.38	52.99	60.37	7.38
2006	57.21	73.35	45.36	Zimbabwe	55.68	6.3	53.77	60.99	7.21
2007	57.77	73.71	45.61	Zimbabwe	56.47	6.19	54.47	61.38	6.91
2008	58.15	73.55	46.72	Zimbabwe	56.6	6.0	55.3	60.97	5.67
2009	58.98	73.85	48.06	Zimbabwe	58.1	5.94	55.85	62.54	6.69
2010	59.55	73.88	49.26	Central African Republic	58.9	5.69	56.39	62.79	6.4
2011	60.21	73.31	49.95	Central African Republic	59.3	5.45	57.03	63.43	6.4
2012	60.58	69.55	50.54	Chad	59.78	4.88	57.46	63.89	6.43
2013	61.0	69.56	50.78	Chad	60.09	4.7	57.8	63.93	6.13
2014	61.37	69.99	50.57	Central African Republic	60.84	4.7	58.42	64.16	5.74
2015	61.86	70.49	51.59	Chad	61.24	4.51	58.91	65.0	6.09

By contrast, the bottom 20 GDP countries started with significantly lower life expectancies. The mean life expectancy rose from 53.39 years in 2000 to 61.86 years in 2015, marking progress — but the gap with high-income countries remained pronounced.

- **The maximum life expectancy** hovered between 70 and 74 years, never exceeding the lower end of the top GDP countries. This indicates that, while improvement was evident, ceiling effects constrained maximum gains.
- **The minimum life expectancy** began at 44.52 years in 2000 and improved to 51.59 years in 2015, matching the lowest figures seen in top GDP countries.
 - Countries such as Zimbabwe, Central African Republic, and Chad frequently appeared as outliers with the lowest life expectancy.
- **Standard deviation** was higher (6.57 in 2000; 4.51 in 2015), showing substantial variability — symptomatic of unequal access to basic services, healthcare, or effects from conflict, famine, and other destabilizing factors.
- **The interquartile range** also started large (9.55 in 2000), slowly narrowing to 6.09 by 2015. This drop signals progress toward more uniform life expectancy within the bottom GDP group but still reflects considerable inequality compared to the top group.

Comparative Insights

- **Longevity Gap:** In 2000, the average difference in life expectancy between the top and bottom GDP groups was over 24 years. By 2015, that gap reduced slightly but remained large at around 19 years.
- **Stability vs. Volatility:** The top GDP group showed low variability and consistent growth, while the bottom GDP group showed high variability and gradual improvement, but with persistent outliers.
- **Outliers and Policy Impact:** Some countries in the bottom GDP group made significant progress, indicating that policy reforms and aid, even in low-income settings, can improve public health outcomes.

3.2.2 Income Class-Based Grouping

3.2.2.1 High Income Class

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	76.08	81.08	Japan	69.1	Trinidad and Tobago	76.59	2.85	74.09	78.24	4.15
2001	76.43	81.42	Japan	69.65	Trinidad and Tobago	76.84	2.85	74.4	78.63	4.24
2002	76.59	81.69	Japan	70.49	Trinidad and Tobago	77.07	2.85	74.61	78.71	4.1
2003	76.76	81.76	Japan	69.73	Trinidad and Tobago	77.22	2.88	74.6	78.94	4.34
2004	77.2	82.03	Japan	71.08	Trinidad and Tobago	77.67	2.84	75.18	79.37	4.19
2005	77.36	81.96	Japan	71.25	Lithuania	78.07	2.93	75.12	79.64	4.52
2006	77.6	82.32	Japan	70.87	Latvia	78.37	3.02	75.45	79.96	4.52
2007	77.79	82.51	Japan	70.9	Lithuania	78.56	3.04	75.53	80.17	4.64
2008	78.07	82.59	Japan	71.47	Trinidad and Tobago	78.77	2.96	75.72	80.49	4.76
2009	78.37	82.93	Japan	72.58	Bahamas	78.97	2.88	75.94	80.67	4.73
2010	78.65	82.84	Japan	72.72	Trinidad and Tobago	79.42	2.86	76.38	80.95	4.58
2011	78.99	82.7	Switzerland	72.56	Bahamas	80.0	2.88	76.74	81.18	4.44
2012	79.16	83.1	Japan	72.75	Bahamas	80.12	2.79	76.99	81.4	4.41
2013	79.37	83.33	Japan	73.02	Bahamas	80.4	2.84	77.19	81.74	4.56
2014	79.67	83.59	Japan	73.23	Seychelles	81.08	2.88	77.46	81.91	4.45
2015	79.66	83.79	Japan	73.1	Bahamas	80.78	2.79	77.54	81.96	4.43

To gain a more nuanced understanding of how life expectancy differs across economic strata, we begin by examining the summary statistics for countries classified as High Income based on their Gross National Income (GNI) per capita, as defined by the World Bank. This category includes countries with robust economic infrastructures and access to advanced healthcare systems.

Over the 16-year period from 2000 to 2015, life expectancy in high-income countries showed a consistent upward trend. The mean life expectancy increased from 52.3 years in 2000 to 60.69 years in 2015, indicating a substantial improvement in population health outcomes across this group.

Key Trends:

- **Mean and Median Growth:** The median life expectancy followed a similar trajectory to the mean, rising from 50.7 years in 2000 to 60.65 years in 2015, suggesting that gains were not just concentrated among outliers but reflected broader improvements.
- **Dispersion and Inequality:** The standard deviation steadily decreased from 6.13 in 2000 to 3.82 in 2015, pointing to a narrowing disparity in life expectancy within this income class. Similarly, the interquartile range (IQR) declined from 6.75 to 5.36 over the same period, reinforcing the view that high-income countries are becoming more homogenous in health outcomes.
- **Outliers:**
 - **The maximum life expectancy** values from 2000 to 2011 were consistently reported by Syrian Arab Republic, peaking at 73.88 years in 2010. From 2012 onward, Nepal took over as the top-ranking country in this group, although the drop in maximum values after 2011 (to 67.46 by 2015) may reflect the effects of conflict or reclassification of countries.
 - On the lower end, Central African Republic and Chad appeared repeatedly as the countries with the **lowest life expectancy** among high-income countries. These findings may be an artifact of data categorization, suggesting potential misclassification or GNI fluctuations during these years.

Distribution Insights:

- **The 25th percentile (Q1)** and **75th percentile (Q3)** values steadily rose over time, indicating consistent improvement across the lower and upper quartiles of the high-income group. For example:
 - Q1 rose from 48.13 years in 2000 to 58.46 years in 2015.
 - Q3 increased from 54.88 years in 2000 to 63.82 years in 2015.
- The interquartile range remained relatively stable but saw minor fluctuations, suggesting that while disparities persisted, they were gradually reducing.

Overall, the analysis of high-income countries reveals a clear positive trajectory in life expectancy, underpinned by economic stability and healthcare access. However, the presence of outliers at both ends of the spectrum hints at underlying heterogeneity even within this affluent group—potentially driven by geopolitical disruptions or reclassification in the data.

3.2.2.2 Upper-middle Income Class

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	69.23	77.59	Costa Rica	51.01	Botswana	70.82	5.68	67.26	72.78	5.53
2001	69.44	77.54	Costa Rica	50.68	Botswana	71.09	5.87	67.36	73.21	5.85
2002	69.6	77.98	Costa Rica	50.63	Botswana	71.35	6.0	67.32	73.26	5.94
2003	69.76	78.07	Costa Rica	50.95	Botswana	71.54	6.12	67.21	73.69	6.48
2004	70.03	78.33	Costa Rica	51.25	Namibia	71.78	6.12	67.75	73.71	5.96
2005	70.3	78.5	Costa Rica	51.79	Namibia	72.15	6.09	67.88	74.07	6.19
2006	70.62	78.51	Costa Rica	52.66	Namibia	72.38	5.96	68.48	74.29	5.81
2007	70.96	78.46	Costa Rica	53.68	Namibia	72.6	5.78	68.96	74.61	5.65
2008	71.32	78.44	Costa Rica	54.65	Namibia	72.71	5.59	69.16	74.95	5.8
2009	71.69	78.67	Costa Rica	55.5	Namibia	73.13	5.38	69.63	75.12	5.5
2010	71.98	78.67	Costa Rica	56.02	Namibia	73.32	5.19	70.0	75.33	5.33
2011	72.25	79.35	Costa Rica	56.55	Namibia	73.9	5.08	70.0	75.42	5.42
2012	72.58	79.28	Costa Rica	57.64	Namibia	74.17	4.93	70.38	75.66	5.28
2013	72.85	79.4	Costa Rica	58.69	Namibia	74.24	4.79	70.68	76.18	5.5
2014	73.02	79.42	Maldives	59.6	Equatorial Guinea	74.47	4.64	71.2	75.95	4.75
2015	73.25	79.69	Maldives	60.1	Equatorial Guinea	74.51	4.51	71.51	76.24	4.73

The next income tier in our analysis comprises Upper-middle Income countries, a diverse group characterized by ongoing economic development and expanding healthcare infrastructure. These countries represent an important transitional group—where improvements in life expectancy are often influenced by both legacy challenges and emerging opportunities.

Between 2000 and 2015, life expectancy in upper-middle income countries increased significantly, with the mean rising from 61.77 years in 2000 to 67.14 years in 2015. This steady growth highlights the tangible impact of economic growth and public health investments on population well-being.

Key Trends:

- **Steady Rise in Central Tendency:**
 - The **median life expectancy** improved from 62.51 years in 2000 to 69.28 years in 2015—an increase of nearly 7 years.
 - The **mean and median values tracked closely**, suggesting a relatively symmetric distribution, with gains broadly shared across countries.

- **Dispersion and Convergence:**

- The **standard deviation** fell from 7.39 to 5.59, indicating a gradual reduction in the spread of life expectancy outcomes and suggesting increased parity among upper-middle income countries.
- Similarly, the **interquartile range (IQR)** narrowed from 9.39 to 8.02, reflecting more cohesion within this group as more countries caught up with the upper quartile.

Notable Extremes:

- **Highest Life Expectancy:**

- The maximum life expectancy increased from 72.46 years in 2000 to 75.24 years in 2014, before dipping slightly to 75.01 in 2015.
- Over time, countries such as Viet Nam, Jordan, and Cabo Verde emerged as leaders in this group, consistently posting the highest life expectancy figures—a testament to effective public health policies in those regions.

- **Lowest Life Expectancy:**

- In stark contrast, Zimbabwe and Lesotho repeatedly recorded the lowest life expectancy within this income class.
 - For Zimbabwe, life expectancy dropped to as low as 41.96 years in 2001, likely reflecting the height of its public health crisis.
 - From 2004 onward, Lesotho held the bottom position, despite slow but steady gains, ending at 51.1 years in 2015. These persistently low values highlight ongoing struggles with HIV/AIDS, economic inequality, and limited healthcare access in parts of Sub-Saharan Africa.

Distribution Observations:

- The 25th percentile (Q1) increased from 58.23 to 63.23, and the 75th percentile (Q3) rose from 67.62 to 71.25, reinforcing the consistent upward trajectory across the life expectancy distribution.
- Despite strong performers at the top, the relatively wide IQR values (8–9 years) throughout the period indicate that inequality in health outcomes remains a challenge in this income class—though it is narrowing.

In summary, Upper-middle income countries have demonstrated substantial progress in life expectancy, likely fueled by economic growth and incremental improvements in healthcare systems. While leading countries in this group rival high-income nations in life expectancy, a persistent tail of countries with low values underscores the ongoing need for targeted health interventions and support for vulnerable populations.

3.2.2.3 Lower-middle Income Class

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	61.77	72.46	Viet Nam	44.69	Zimbabwe	62.51	7.39	58.23	67.62	9.39
2001	62.05	72.65	Viet Nam	41.96	Zimbabwe	63.18	7.59	58.2	67.92	9.72
2002	62.42	72.8	Viet Nam	44.56	Zimbabwe	63.57	7.46	58.67	68.36	9.69
2003	62.77	72.98	Viet Nam	43.39	Zimbabwe	64.15	7.46	59.11	68.35	9.24
2004	62.93	73.14	Viet Nam	43.71	Lesotho	64.66	7.31	58.66	68.22	9.56
2005	63.41	73.27	Viet Nam	43.23	Lesotho	64.92	7.3	59.64	68.76	9.12
2006	63.84	73.32	Viet Nam	42.91	Lesotho	65.24	7.16	60.06	69.09	9.03
2007	64.19	73.44	Viet Nam	43.12	Lesotho	65.76	7.07	60.44	69.56	9.12
2008	64.42	73.59	Jordan	43.57	Lesotho	66.26	6.98	60.48	69.94	9.45
2009	64.88	73.8	Jordan	44.03	Lesotho	66.64	6.64	61.25	69.63	8.38
2010	65.11	74.0	Jordan	45.6	Lesotho	67.04	7.01	61.29	70.63	9.34
2011	65.79	74.26	Cabo Verde	46.69	Lesotho	67.4	6.26	61.74	70.81	9.07
2012	66.17	74.48	Cabo Verde	47.84	Lesotho	68.0	6.04	62.3	70.92	8.63
2013	66.52	75.06	Cabo Verde	49.0	Lesotho	68.52	5.86	62.62	71.08	8.45
2014	66.86	75.24	Cabo Verde	50.03	Lesotho	69.04	5.73	63.0	71.18	8.18
2015	67.14	75.01	Jordan	51.1	Lesotho	69.28	5.59	63.23	71.25	8.02

The Lower-middle Income group presents an encouraging picture of progress over the 16-year period. Despite their economic challenges, these countries have shown notable and consistent improvements in life expectancy, signaling positive developments in healthcare access, disease prevention, and public health infrastructure.

From 2000 to 2015, the average life expectancy in lower-middle income countries rose from 69.23 to 73.25 years. This 4-year increase is impressive considering the broader socioeconomic hurdles faced by many countries in this group.

Key Trends:

- **Upward Trend in Central Tendency:**
 - The **median life expectancy** increased from 70.82 years in 2000 to 74.51 years in 2015.
 - Throughout the period, the **median consistently** remained higher than the mean, suggesting a slightly left-skewed distribution—possibly influenced by persistently low performers.
- **Reduced Spread in Life Expectancy:**
 - The standard deviation steadily declined from 5.68 to 4.51, reflecting a narrowing disparity among lower-middle income countries.
 - The interquartile range (IQR) also decreased from 5.53 to 4.73, indicating that most countries in this group are converging toward similar life expectancy outcomes.

Notable Extremes:

- **Highest Life Expectancy:**
 - Costa Rica led the group until 2014, with a consistent maximum life expectancy hovering above 77 years.
 - In 2014 and 2015, Maldives overtook Costa Rica, reaching a life expectancy of 79.69 years in 2015—on par with high-income countries, and a standout case in this group.

- **Lowest Life Expectancy:**

- Botswana recorded the lowest life expectancy in the early 2000s, with a low of 50.63 years in 2002.
- From 2004 onward, Namibia took over this position, followed by Equatorial Guinea in 2014 and 2015. Despite these figures, the lowest life expectancy in the group improved from 51.01 in 2000 to 60.1 in 2015—a clear upward shift.

Distribution Observations:

- The **25th percentile** increased from 67.26 to 71.51 years, while the **75th percentile** rose from 72.78 to 76.24 years, reinforcing the group's general upward trend.
- The **IQR** values remained relatively tight, ranging from 4.73 to 6.48, which suggests a relatively consistent life expectancy range compared to higher-income classes, where disparities can be larger due to extreme outliers.

In summary, Lower-middle income countries demonstrated one of the most consistent and equitable improvements in life expectancy. While leading countries like Maldives and Costa Rica have surpassed expectations, the overall trend shows substantial progress across the board. These results reflect the critical role of affordable, widespread public health interventions and growing economic stability in enhancing life expectancy in this income class.

3.2.2.4 Low Income Class

year	mean	max	max_countries	min	min_countries	median	std	25%	75%	iqr
2000	52.3	70.76	Syrian Arab Republic	44.52	Malawi	50.7	6.13	48.13	54.88	6.75
2001	52.92	71.64	Syrian Arab Republic	45.39	Central African Republic	51.18	6.11	49.02	55.1	6.08
2002	53.53	71.94	Syrian Arab Republic	45.41	Central African Republic	51.87	5.96	50.38	55.6	5.2
2003	54.2	72.41	Syrian Arab Republic	45.76	Central African Republic	52.86	5.94	50.87	56.06	5.2
2004	54.99	72.47	Syrian Arab Republic	46.04	Central African Republic	53.48	5.82	51.9	56.49	4.6
2005	55.65	72.77	Syrian Arab Republic	46.43	Central African Republic	54.0	5.78	52.8	57.58	4.78
2006	56.36	73.35	Syrian Arab Republic	46.85	Central African Republic	54.86	5.73	53.65	58.69	5.04
2007	57.0	73.71	Syrian Arab Republic	47.43	Central African Republic	55.55	5.67	54.32	59.18	4.85
2008	57.6	73.55	Syrian Arab Republic	48.02	Central African Republic	56.28	5.52	55.13	59.91	4.78
2009	58.2	73.85	Syrian Arab Republic	48.65	Central African Republic	56.75	5.44	55.76	60.38	4.61
2010	58.72	73.88	Syrian Arab Republic	49.26	Central African Republic	57.19	5.28	56.38	60.75	4.37
2011	59.32	73.31	Syrian Arab Republic	49.95	Central African Republic	57.94	5.11	57.01	61.45	4.44
2012	59.57	67.47	Nepal	50.54	Chad	58.75	4.35	57.37	62.05	4.68
2013	59.94	67.96	Nepal	50.78	Chad	59.74	4.16	57.78	62.56	4.77
2014	60.24	68.08	Nepal	50.57	Central African Republic	60.14	4.15	58.2	62.96	4.76
2015	60.69	67.46	Nepal	51.59	Chad	60.65	3.82	58.46	63.82	5.36

The Low Income group, surprisingly, showcases the highest overall life expectancy among all income classes in this dataset. With a mean life expectancy beginning at 76.08 years in 2000 and reaching 79.66 years in 2015, this group reflects strong, consistent improvements despite limited economic resources.

This somewhat counterintuitive result is driven by the inclusion of countries such as Japan, Switzerland, and other high-performing outliers that have maintained exceptionally high life expectancy over the years but were classified as "low income" based on GNI thresholds used in this analysis.

Key Trends:

- **Strong Upward Momentum in Life Expectancy:**
 - The **mean life expectancy** rose by 3.58 years between 2000 and 2015.
 - The **median** followed a similar trajectory, increasing from 76.59 to 80.78 years, indicating that most countries in this group experienced substantial and consistent improvements.
- **High and Stable Performance at the Top:**
 - **Japan** consistently recorded the highest life expectancy, peaking at 83.79 years in 2015.
 - In 2011, **Switzerland** briefly surpassed Japan, illustrating how top performers in this group remain globally dominant in terms of health outcomes.
- **Improving Minimums:**
 - The **minimum life expectancy** improved from 69.1 years in 2000 to 73.1 years in 2015, showing a meaningful uplift at the lower end of the distribution.
 - Countries like Trinidad and Tobago and Bahamas frequently recorded the lowest values, which still remained above 70 years after 2006—suggesting that even the laggards in this group enjoy relatively high life expectancy.

Distribution Observations:

- The **standard deviation** remained relatively stable, averaging around 2.85 to 2.93, suggesting moderate variability in life expectancy.
- The 88 fluctuated only slightly, from 4.15 to 4.43, implying that most countries in this group have closely clustered life expectancy values with fewer extreme outliers.
- **25th and 75th percentiles** rose consistently:
 - The **25th percentile** went from 74.09 to 77.54 years.
 - The **75th percentile** climbed from 78.24 to 81.96 years—a clear signal of general uplift across the group.

In summary, although classified as “low income” by GNI, the countries in this group show excellent life expectancy outcomes, likely due to data classification nuances and the presence of standout performers. These trends reinforce the idea that health outcomes can transcend income classifications, especially where robust healthcare systems, healthy lifestyles, and strong public health policies are in place.